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Inventor(s)	Masada; Tatsuya et al.

Electrode assembly cutting device and separator cutting device

Abstract

A device for manufacturing a laminated electrode comprises: a negative electrode cutting drum that forms a negative electrode sheet by cutting a negative electrode single sheet to a first width and conveys the negative electrode sheet; a negative electrode heating drum that heats the negative electrode sheet; a positive electrode cutting drum that forms a positive electrode sheet by cutting a positive electrode single sheet to a second width and conveys the positive electrode sheet; a positive electrode heating drum that heats the positive electrode sheet; and a bonding drum at which the negative electrode sheet is arranged on a first separator single sheet, a second separator single sheet is arranged on the negative electrode sheet, the positive electrode sheet is arranged on the second separator single sheet, and the sheets are bonded. The negative and positive electrode cutting drums include: a plurality of holding heads and pairs of blades.

Inventors:	Masada; Tatsuya (Hyogo, JP), Inagaki; Kenji (Hyogo, JP), Maruyama; Masahide (Nara, JP)
Applicant:	Panasonic Corporation (Kadoma, JP); SANYO Electric Co., Ltd. (Daito, JP)
Family ID:	1000008821090
Assignee:	PANASONIC HOLDINGS CORPORATION (Osaka, JP); PANASONIC ENERGY CO., LTD. (Osaka, JP)
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Primary Examiner: MacFarlane; Evan H

Attorney, Agent or Firm: WHDA, LLP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to an electrode assembly cutting device and a separator cutting device.

BACKGROUND ART

(2) Since a large capacity, low internal resistance and a high heat radiation performance are requested in an in-vehicle lithium ion secondary battery or the like for example, a laminate type battery which can satisfy the requests has been developed. The battery is in a form that a positive electrode, a separator and a negative electrode are alternately laminated, the respective electrodes are connected to a metal terminal called a tab and put in a container configured by an aluminum laminate film, an electrolyte is injected and sealing is performed.

(3) Patent Literature 1 describes that there are: an electrode support section that receives an electrode supplied by a conveying device and supports the electrode; a circulation member which has a loop shape extending in a vertical direction, and on the outer peripheral surface of which the electrode support section is attached; a lamination unit that is disposed on a side opposite to the conveying device with the circulation member interposed therebetween, and includes a plurality of stages of lamination sections where the electrode is laminated; an extrusion section that simultaneously extrudes the electrodes supported by the plurality of electrode support sections toward the plurality of stages of lamination sections; and a control section that controls circulation and elevation of the circulation member and an operation of the extrusion section, and the control section controls the operation of the extrusion section so as to extrude the electrodes toward the lamination sections at a speed lower than a conveying speed of the electrodes by the conveying device.

(4) Patent Literature 2 describes that a device which manufactures a laminated electrode assembly for which a positive electrode and a negative electrode are alternately laminated holding a separator therebetween comprises: a winding drum having an outer peripheral surface around which a continuously delivered separator sheet can be wound; and an electrode supply unit which supplies the electrode successively at intervals such that the positive electrode and the negative electrode can be switched, to a valley formed between a separator sheet SS wound around the outer peripheral surface and a separator sheet to be wound around the outer peripheral surface.

(5) Patent Literature 3 describes that raw materials are a continuous positive electrode material, a continuous negative electrode material and a continuous separator material for which positive electrodes, negative electrodes and separators are continuously formed via easy-to-cut breaking

lines respectively, there are roughly cylindrical winding means that wind a continuous battery material formed by piling up the respective raw materials in the order of the continuous separator material, the continuous positive electrode material, the continuous separator material and the continuous negative electrode material or in the order of the continuous separator material, the continuous negative electrode material, the continuous separator material and the continuous positive electrode material while the respective breaking lines are made to coincide with one another, pressing means that press the wound continuous battery material to a side peripheral surface of the winding means, and cutting means that cut the continuous battery material at each breaking line after the continuous battery material is wound around the winding means for a required number of laminations, and for the cutting means, a part of the side peripheral surface of the winding means in a circumferential direction is projected in a radial direction.

CITATION LIST

Patent Literature

(6) PATENT LITERATURE 1: International Publication No. WO 2017/131027 PATENT

LITERATURE 2: Japanese Unexamined Patent Application Publication No. 2012-199211 PATENT

LITERATURE 3: Japanese Unexamined Patent Application Publication No. 2011-86508

SUMMARY

(7) In order to efficiently manufacture a laminated electrode assembly formed by alternately laminating a positive electrode, a separator and a negative electrode, it is needed to supply a belt-like positive electrode body and a belt-like negative electrode body to a drum, obtain a positive electrode plate and a negative electrode plate by cutting the belt-like positive electrode body and negative electrode body into desired sizes respectively, and successively laminate them at desired positions.

(8) It is an object of the present disclosure to provide a technology capable of efficiently cutting a rectangular electrode assembly configuring a laminated electrode assembly when manufacturing the laminated electrode assembly.

(9) One aspect of the present disclosure comprises: a plurality of holding heads arranged in a circumferential direction of a drum and rotated around a drum center axis holding a rectangular electrode assembly; and cutting means provided in each of the plurality of holding heads and configured to cut the rectangular electrode assembly in a predetermined width, and the cutting means comprise a pair of blades arranged in a thickness direction of the rectangular electrode assembly so as to hold the rectangular electrode assembly therebetween.

(10) In another aspect of the present disclosure, the pair of blades maintains a distance so as not to overlap with each other in the thickness direction of the rectangular electrode assembly and cuts the rectangular electrode assembly.

(11) In a further aspect of the present disclosure, a distance L of the pair of blades in the thickness direction of the rectangular electrode assembly and a thickness d of the electrode assembly satisfy $d > L \geq 0$.

(12) In a still further aspect of the present disclosure, the rectangular electrode assembly comprises a current collector and an active material layer formed on at least one surface of the current collector, and the pair of blades cuts the active material layer and does not cut the current collector.

(13) In a yet further aspect of the present disclosure, the pair of blades forms a cut without cutting the current collector.

(14) In a yet still further aspect of the present disclosure, the current collector is cut by tensile stress by rotation of the holding heads.

(15) In a yet still further aspect of the present disclosure, the rectangular electrode assembly comprises a current collector and an active material layer formed on at least one surface of the current collector, the current collector has a cut surface by tensile stress, and the active material layer has a cut surface by shearing stress.

(16) In a yet still further aspect of the present disclosure, the cutting means cut the rectangular

electrode assembly by reciprocating in a width direction of the rectangular electrode assembly, which is a direction roughly orthogonal to the circumferential direction of the drum.

(17) In a yet still further aspect of the present disclosure, a moving surface during forward movement and a moving surface during return movement in the reciprocation are different from each other.

(18) In a yet still further aspect of the present disclosure, the cutting means cut the rectangular electrode assembly by being moved forward in the width direction of the rectangular electrode assembly, which is the direction roughly orthogonal to the circumferential direction of the drum, the holding head holding the cut rectangular electrode assembly is moved in a direction of separating from the holding head holding the rectangular electrode assembly, and the cutting means are moved back in the width direction of the rectangular electrode assembly, which is the direction roughly orthogonal to the circumferential direction of the drum, after being moved in the direction of separating from the rectangular electrode assembly.

(19) In a yet still further aspect of the present disclosure, each blade of the pair of blades is a rotary round blade.

(20) In a yet still further aspect of the present disclosure, each blade of the pair of blades is a rotary round blade with a flat portion formed at one part.

(21) In a yet still further aspect of the present disclosure, the pair of blades cuts the rectangular electrode assembly by round blade portions facing each other during forward movement in the reciprocation, and does not come into contact with the rectangular electrode assembly while the flat portions face each other during return movement.

(22) A yet still further aspect of the present disclosure is a separator cutting device comprising: a plurality of holding heads arranged in a circumferential direction of a drum and rotated around a drum center axis holding a belt-like separator; and cutting means provided in each of the plurality of holding heads and configured to cut the belt-like separator in a predetermined width, the cutting means comprise a pair of blades arranged in a thickness direction of the belt-like separator so as to hold the belt-like separator therebetween, and the cutting means cut the belt-like separator by reciprocating in a width direction of the belt-like separator, which is a direction roughly orthogonal to the circumferential direction of the drum.

(23) In a yet still further aspect of the present disclosure, a moving surface during forward movement and a moving surface during return movement in the reciprocation are different from each other.

(24) In a yet still further aspect of the present disclosure, the cutting means cut the belt-like separator by being moved forward in the width direction of the belt-like separator, which is the direction roughly orthogonal to the circumferential direction of the drum, the holding head holding the cut separator is moved in a direction of separating from the holding head holding the belt-like separator, and the cutting means are moved back in the width direction of the belt-like separator, which is the direction roughly orthogonal to the circumferential direction of the drum, after being moved in the direction of separating from the belt-like separator.

(25) In a yet still further aspect of the present disclosure, each blade of the pair of blades is a rotary round blade.

(26) In a yet still further aspect of the present disclosure, each blade of the pair of blades is a rotary round blade with a flat portion formed at one part.

(27) In a yet still further aspect of the present disclosure, the pair of blades cuts the belt-like separator by round blade portions facing each other during the forward movement in the reciprocation, and does not come into contact with the belt-like separator while the flat portions face each other during the return movement.

(28) One aspect of the present disclosure makes it possible to efficiently cut a rectangular electrode assembly configuring a laminated electrode assembly.

(29) In addition, another aspect of the present disclosure further makes it possible to suppress damages of a cut surface of the rectangular electrode assembly.

Description

BRIEF DESCRIPTION OF DRAWING

- (1) FIG. 1 is a conceptual configuration diagram of a manufacturing device of an embodiment.
- (2) FIG. 2 is a configuration perspective view of the manufacturing device of the embodiment.
- (3) FIG. 3 is an explanatory diagram illustrating a manufacturing method of a 4-layer laminated body of the embodiment.
- (4) FIG. 4 is a configuration diagram of the 4-layer laminated body of the embodiment.
- (5) FIG. 5 is an explanatory diagram illustrating a manufacturing method of a 3-layer laminated body of the embodiment.
- (6) FIG. 6 is a configuration diagram of the 3-layer laminated body of the embodiment.
- (7) FIG. 7 is a lamination explanatory diagram of the 3-layer laminated body and the 4-layer laminated body of the embodiment.
- (8) FIG. 8 is a cutting explanatory diagram of a separator of the embodiment.
- (9) FIG. 9 is a generation order explanatory diagram of the 3-layer laminated body and the 4-layer laminated body of the embodiment.
- (10) FIG. 10 is a configuration diagram of a laminated electrode assembly of the embodiment.
- (11) FIG. 11 is a configuration perspective view of a positive electrode cutting head of the embodiment.
- (12) FIG. 12 is a cutting explanatory diagram of a positive electrode single plate of a comparative example.
- (13) FIG. 13 is a cutting explanatory diagram (part 1) of the positive electrode single plate of the embodiment.
- (14) FIG. 14 is a cutting explanatory diagram (part 2) of the positive electrode single plate of the embodiment.
- (15) FIG. 15 is a reciprocation explanatory diagram of a round blade of the embodiment.
- (16) FIG. 16 is a top view and a side view of the positive electrode single plate of the embodiment.
- (17) FIG. 17 is a cut surface explanatory diagram of the positive electrode single plate of the embodiment.
- (18) FIG. 18 is a cut surface explanatory diagram of the positive electrode single plate of the comparative example.
- (19) FIG. 19 is an explanatory diagram illustrating another blade shape of the embodiment.
- (20) FIG. 20 is a reciprocation explanatory diagram of another blade of the embodiment.
- (21) FIG. 21 is a supply explanatory diagram of the positive electrode single plate of the embodiment.
- (22) FIG. 22 is a partially enlarged view of FIG. 21.
- (23) FIG. 23A is an explanatory diagram of a negative electrode heating drum and a bonding drum of the embodiment.
- (24) FIG. 23B is a perspective view of a pressurizing force adjusting mechanism of the embodiment.
- (25) FIG. 23C is a rear elevation of the pressurizing force adjusting mechanism of the embodiment.
- (26) FIG. 24 is a bonding explanatory diagram of a negative electrode plate of the comparative example.
- (27) FIG. 25 is a bonding explanatory diagram of a negative electrode plate of the embodiment.
- (28) FIG. 26 is a pressurizing range explanatory diagram for the time of bonding the negative electrode plate of the embodiment.

- (29) FIG. **27** is a pressurizing range explanatory diagram for the time of bonding a separator **S2** of the embodiment.
- (30) FIG. **28** is a pressurizing range explanatory diagram for the time of bonding a negative electrode plate of the embodiment.
- (31) FIG. **29** is a configuration perspective view of a laminating drum of the embodiment.
- (32) FIG. **30** is an operation explanatory diagram (part 1) of the laminating drum of the embodiment.
- (33) FIG. **31** is an operation explanatory diagram (part 2) of the laminating drum of the embodiment.
- (34) FIG. **32** is an operation explanatory diagram (part 3) of the laminating drum of the embodiment.
- (35) FIG. **33** is an operation explanatory diagram (part 4) of the laminating drum of the embodiment.
- (36) FIG. **34** is a configuration perspective view of a laminating head of the embodiment.
- (37) FIG. **35** is a graphical representation illustrating a position change of the laminating head of the embodiment.
- (38) FIG. **36** is a configuration perspective view of a laminating stage of the embodiment.
- (39) FIG. **37** is a diagram illustrating a claw arrangement of the laminating stage of the embodiment.
- (40) FIG. **38** is a claw operation explanatory diagram of the laminating stage of the embodiment.
- (41) FIG. **39** is a plan view illustrating another claw arrangement of the laminating stage of the embodiment.
- (42) FIG. **40** is a linked operation explanatory diagram (part 1) of the laminating head and claws of the embodiment.
- (43) FIG. **41** is a linked operation explanatory diagram (part 2) of the laminating head and the claws of the embodiment.
- (44) FIG. **42** is a linked operation explanatory diagram (part 3) of the laminating head and the claws of the embodiment.
- (45) FIG. **43** is a linked operation explanatory diagram (part 4) of the laminating head and the claws of the embodiment.
- (46) FIG. **44** is a linked operation explanatory diagram (part 5) of the laminating head and the claws of the embodiment.
- (47) FIG. **45** is a conceptual configuration diagram of a modification 1.
- (48) FIG. **46** is a lamination explanatory diagram of the modification 1.
- (49) FIG. **47** is another lamination explanatory diagram of the modification 1.
- (50) FIG. **48** is a conceptual configuration diagram of a modification 2.
- (51) FIG. **49** is another conceptual configuration diagram of the modification 2.
- (52) FIG. **50** is a conceptual configuration diagram of a modification 3.
- (53) FIG. **51** is a conceptual configuration diagram of a modification 4.

DESCRIPTION OF EMBODIMENTS

(54) Hereinafter, a laminated electrode assembly manufacturing device and manufacturing method relating to one aspect of the present disclosure will be described. However, the embodiment described hereinafter is an example and the present disclosure is not limited thereto.

(55) FIG. **1** illustrates a conceptual diagram of a laminated electrode assembly manufacturing device in the present embodiment. The manufacturing device of the embodiment is a continuous drum type manufacturing device for which a plurality of drums are combined, and comprises a negative electrode cutting drum **10**, a negative electrode heating drum **12**, a positive electrode cutting drum **14**, a positive electrode heating drum **16**, a bonding drum **18**, a separator cutting drum **20** and a laminating drum **22**.

(56) The negative electrode cutting drum **10** is a first electrode cutting drum, has a first radius, and

is rotated around a central axis at a first angular velocity. To the negative electrode cutting drum **10**, a belt-like negative electrode single plate N is supplied as a first electrode single plate. The negative electrode single plate N is a negative electrode. The negative electrode single plate N is configured by a negative electrode current collector, and a negative electrode active material layer formed on the negative electrode current collector. The negative electrode active material layer may be formed on one surface of the negative electrode current collector, or may be formed on both surfaces. In the description below, it is assumed that the negative electrode active material layer is formed on both surfaces of the negative electrode current collector. The negative electrode active material layer includes a negative electrode active material and a binding material.

(57) For both of the negative electrode current collector and the negative electrode active material layer, known materials can be used, and are exemplified as follows for a lithium ion secondary battery.

(58) As the negative electrode current collector, foil of a metal which is stable in a potential range of the negative electrode, and a film for which the metal is arranged on a surface layer or the like can be used. As the negative electrode current collector, a porous body such as a mesh body, a punching sheet or an expand metal of the metal may be used. As the material of the negative electrode current collector, copper, copper alloy, aluminum, aluminum alloy, stainless steel, nickel or the like can be used. A thickness of the negative electrode current collector is preferably 3 μm or more and 50 μm or less for example, from viewpoints of a current collection property, mechanical strength or the like. The negative electrode single plate N can be manufactured by applying negative electrode mixture slurry including the negative electrode active material, the binding material and a dispersion medium onto the negative electrode current collector, drying a coating film, then performing rolling and forming the negative electrode active material layer on one surface or both surfaces of the negative electrode current collector for example. The negative electrode active material layer may include an arbitrary component such as a conductive agent as needed. The thickness of the negative electrode active material layer is not limited in particular, and is 10 μm or more and 100 μm or less for example.

(59) In the case of the lithium ion secondary battery, the negative electrode active material is not limited in particular as long as it is a material capable of occluding/releasing lithium ions. The material configuring the negative electrode active material may be a non-carbon-based material, may be a carbon material, or may be the combination thereof.

(60) Examples of the non-carbon-based material are lithium metal, alloy including a lithium element and a metal compound such as metal oxide, metal sulfide or metal nitride containing lithium. Examples of the alloy containing the lithium element are lithium aluminum alloy, lithium tin alloy, lithium lead alloy and lithium silicon alloy. An example of the metal oxide containing the lithium is the metal oxide containing the lithium and titanium, tantalum, niobium or the like, and lithium titanate ($\text{Li}_{0.4}\text{Ti}_{0.5}\text{O}_{1.2}$ or the like) is preferable. Examples of the carbon material used as the negative electrode active material are graphite and hard carbon. Among them, the graphite is preferable since a capacity is high and an irreversible capacity is small. The graphite is a general term of the carbon material having a graphite structure, and includes natural graphite, artificial graphite, expanded graphite, graphitized mesophase carbon particles or the like. In the case of using the graphite as the negative electrode active material, it is preferable to coat the surface of the negative electrode active material layer with a coating film in order to lower activity to reductive decomposition of an aqueous electrolyte. For the negative electrode active materials, one kind may be used alone, or two or more kinds may be used together. As the binding material included in the negative electrode active material layer, a fluorine-based polymer, a rubber-based polymer or the like may be used, and a styrene-butadiene copolymer (SBR) or the modified product or the like may be used.

(61) The negative electrode cutting drum **10** comprises a plurality of electrode cutting heads arranged in a circumferential direction of the drum. The plurality of electrode cutting heads

comprise an outer peripheral surface for sucking and holding the negative electrode single plate N, and cutting means. The cutting means are, for example, a blade moved in a direction roughly orthogonal to the circumferential direction of the outer peripheral surface. The supplied negative electrode single plate N is sucked and held on the outer peripheral surface and rotated. The electrode cutting heads suck and hold the negative electrode single plate N so that they are also referred to as holding heads. A gap is formed between the plurality of electrode cutting heads, and by movement of the blade loaded on the electrode cutting head in the direction roughly orthogonal to the circumferential direction at the gap, the negative electrode single plate N sucked and held on the outer peripheral surface is cut to have a predetermined width (first width) by the blade.

(62) The plurality of electrode cutting heads are each rotated around a common central axis of the negative electrode cutting drum **10**, and the individual electrode cutting head is driven by a motor in the circumferential direction of the drum independent of the other electrode cutting heads. For example, when the two electrode cutting heads adjacent in the circumferential direction are an electrode cutting head 'a' and an electrode cutting head 'b', the electrode cutting head 'a' and the electrode cutting head 'b' are rotated around the common central axis of the drum at a fixed speed, and a mutual relative speed is changed for every predetermined section on a circumference of the negative electrode cutting drum **10**. For example, the electrode cutting head 'a' and the electrode cutting head 'b' are both rotated at the fixed speed and the relative speed is 0 at a certain timing, but the electrode cutting head 'a' is accelerated in a direction of separating from the following electrode cutting head 'b' and the relative speed becomes finite at a different timing. By such independent drive of the electrode cutting head, a cutting position of the negative electrode single plate N by the blade loaded on the electrode cutting head can be adjusted, and a position of a negative electrode plate generated by cutting can be adjusted. A moving speed of the electrode cutting head can be achieved by using a motor or the like corresponding to each electrode cutting head.

(63) The negative electrode cutting drum **10** may comprise various kinds of cameras, and the position of the negative electrode single plate N before cutting may be monitored and also the positions of the plurality of negative electrode plates generated by cutting may be monitored by the cameras. The negative electrode cutting drum **10** sucks, holds, rotationally conveys the supplied negative electrode single plate N, cuts the negative electrode single plate N at a position **11** in FIG. **1**, and generates the negative electrode plate. The electrode cutting head rotated while sucking and holding the negative electrode single plate N is rotated while sucking and holding the negative electrode single plate N to the position **11**, and cuts the negative electrode single plate N by the loaded blade at a point of time of reaching the position **11**. The negative electrode plate of the first width generated by cutting is rotationally conveyed while being kept sucked and held on the outer peripheral surface of each electrode cutting head.

(64) The negative electrode heating drum **12** is a first electrode heating drum, and is arranged adjacently to the negative electrode cutting drum **10** so as to be close to the negative electrode cutting drum **10**. The negative electrode heating drum **12** has a second radius, and is rotated around the central axis at a second angular velocity. The second radius of the negative electrode heating drum **12** may be same as the first radius of the negative electrode cutting drum **10** or may be different. The second angular velocity of the negative electrode heating drum **12** is different from the first angular velocity of the negative electrode cutting drum **10**. Specifically, the second angular velocity of the negative electrode heating drum **12** is set such that the linear velocity is roughly same as the linear velocity of the bonding drum **18** to be described later. As one example, the second radius and the first radius are the same, and setting is performed to be the second angular velocity > the first angular velocity. In this case, the linear velocities of the negative electrode cutting drum **10** and the negative electrode heating drum **12** are different, and are the linear velocity of the negative electrode heating drum **12** > the linear velocity of the negative electrode cutting drum **10**. Accordingly, the electrode cutting head of the negative electrode cutting drum **10**

is temporarily accelerated to be roughly same as the linear velocity of the negative electrode heating drum **12** in front of a position close to the negative electrode heating drum **12**, and turns the relative speed with the negative electrode heating drum **12** to roughly zero. The electrode cutting head of the negative electrode cutting drum **10** discharges the sucked and held negative electrode plate to a side of the negative electrode heating drum **12** at the timing when the relative speed becomes roughly zero. The electrode cutting head of the negative electrode cutting drum **10** is switched to the speed before acceleration after discharging the sucked and held negative electrode plate.

(65) The negative electrode heating drum **12** sucks and holds the negative electrode plate discharged from the negative electrode cutting drum **10**, and heats (preliminarily heats) the negative electrode plate by a built-in heater. The figure illustrates that the negative electrode plate is heated at a position **13**. The heating (preliminary heating) process is for heat-bonding a separator and the negative electrode plate in a subsequent bonding process. A location of heating by the negative electrode heating drum **12** is not limited to a specific position (for example, the position **13**). The negative electrode heating drum **12** may be in a heated state at all times while the drum is rotated.

(66) The positive electrode cutting drum **14** is a second electrode cutting drum, has a third radius, and is rotated around the central axis at a third angular velocity. To the positive electrode cutting drum **14**, a belt-like positive electrode single plate P is supplied as a second electrode single plate. The positive electrode single plate P is a rectangular electrode assembly. The positive electrode single plate P is configured by a positive electrode current collector, and a positive electrode active material layer formed on the positive electrode current collector. The positive electrode active material layer may be formed on one surface of the positive electrode current collector, or may be formed on both surfaces. In the description below, it is assumed that the positive electrode active material layer is formed on both surfaces of the positive electrode current collector. The positive electrode active material layer includes a positive electrode active material and a binding material.

(67) For both of the positive electrode current collector and the positive electrode active material layer, known materials can be used, and are exemplified as follows.

(68) As the positive electrode current collector, foil of a metal which is stable in a potential range of the positive electrode, and a film for which the metal is arranged on a surface layer or the like can be used. As the positive electrode current collector, a porous body such as a mesh body, a punching sheet or an expand metal of the metal may be used. As the material of the positive electrode current collector, the stainless steel, the aluminum, the aluminum alloy, titanium or the like can be used. A thickness of the positive electrode current collector is preferably 3 μm or more and 50 μm or less for example, from the viewpoints of the current collection property, the mechanical strength or the like. The positive electrode single plate can be obtained by forming the positive electrode active material layer on the positive electrode current collector by applying/drying positive electrode mixture slurry including the positive electrode active material, the conductive material, the binding material or the like on the positive electrode current collector, and rolling the positive electrode active material layer, for example. The thickness of the positive electrode active material layer is not limited in particular, and is 10 μm or more and 100 μm or less for example.

(69) The positive electrode active material is lithium transition metal oxide containing lithium (Li) and a transition metal element such as cobalt (Co), manganese (Mn) and nickel (Ni). An example of the conductive material included in the positive electrode active material layer is carbon powder such as carbon black, acetylene black, Ketjen black, graphite or the like. For them, one kind may be used alone or two or more kinds may be combined and used. Examples of the binding material included in the positive electrode active material layer are the fluorine-based polymer, the rubber-based polymer and the like. Examples of the fluorine-based polymer are polytetrafluoroethylene (PTFE), polyvinylidene fluoride (PVDF) and the modified product or the like, and examples of the rubber-based polymer are an ethylene-propylene-isoprene copolymer and an ethylene-propylene-

butadiene copolymer. For them, one kind may be used alone or two or more kinds may be combined and used.

(70) The positive electrode cutting drum **14** comprises a plurality of electrode cutting heads arranged in the circumferential direction of the drum. The electrode cutting head comprises an outer peripheral surface for sucking and holding the positive electrode single plate P, and cutting means. The cutting means are, for example, a blade moved in the direction roughly orthogonal to the circumferential direction of the outer peripheral surface. The supplied positive electrode single plate P is sucked and held on the outer peripheral surface and rotated. The electrode cutting heads suck and hold the positive electrode single plate P so that they are also referred to as holding heads. A gap is formed in the circumferential direction between the plurality of electrode cutting heads, and by the movement of the blade loaded on the electrode cutting head in the direction roughly orthogonal to the circumferential direction at the gap, the positive electrode single plate P sucked and held on the outer peripheral surface is cut to have a predetermined width (second width) by the blade. Similarly to the electrode cutting heads of the negative electrode cutting drum **10**, the plurality of electrode cutting heads are each rotated around the common central axis of the positive electrode cutting drum **14**, and the individual electrode cutting head is driven by a motor in the circumferential direction of the drum independent of the other electrode cutting heads. For example, when the two electrode cutting heads adjacent in the circumferential direction are an electrode cutting head 'a' and an electrode cutting head 'b', the electrode cutting head 'a' and the electrode cutting head 'b' are rotated around the common central axis of the drum at the fixed speed, and the mutual relative speed is changed for every predetermined section on the circumference of the positive electrode cutting drum **14**. For example, the electrode cutting head 'a' and the electrode cutting head 'b' are both rotated at the fixed speed and the relative speed is 0 at a certain timing, but the electrode cutting head 'a' is accelerated in the direction of separating from the following electrode cutting head 'b' and the relative speed becomes finite at a different timing. By such independent drive of the electrode cutting head, the cutting position of the positive electrode single plate P by a round blade loaded on the electrode cutting head can be adjusted, and the position of a positive electrode plate generated by cutting can be adjusted. The moving speed of the electrode cutting head can be achieved by using the motor or the like corresponding to each electrode cutting head.

(71) The positive electrode cutting drum **14** may comprise various kinds of cameras, and the position of the positive electrode single plate P before cutting may be monitored and also the positions of the plurality of positive electrode plates generated by cutting may be monitored by the cameras. The positive electrode cutting drum **14** sucks, holds and rotationally conveys the supplied positive electrode single plate P, cuts the positive electrode single plate P at a position **15** in FIG. **1**, and generates the positive electrode plate. The electrode cutting head rotated while sucking and holding the positive electrode single plate P is rotated while sucking and holding the positive electrode single plate P to the position **15**, and cuts the positive electrode single plate P by the loaded blade at the point of time of reaching the position **15**. The positive electrode plate generated by cutting is rotationally conveyed while being kept sucked and held on the outer peripheral surface of each electrode cutting head.

(72) The positive electrode heating drum **16** is a second electrode heating drum, and is arranged adjacently to the positive electrode cutting drum **14** so as to be close to the positive electrode cutting drum **14**. The positive electrode heating drum **16** has a fourth radius, and is rotated around the central axis at a fourth angular velocity. The fourth radius of the positive electrode heating drum **16** may be the same as the third radius of the positive electrode cutting drum **14** or may be different. The fourth angular velocity of the positive electrode heating drum **16** is different from the third angular velocity of the positive electrode cutting drum **14**. Specifically, the fourth angular velocity of the positive electrode heating drum **16** is set such that the linear velocity is roughly the same as the linear velocity of the bonding drum **18** to be described later. As one example, the fourth

radius and the third radius are the same, and setting is performed to be the fourth angular velocity>the third angular velocity. In this case, the linear velocities of the positive electrode cutting drum **14** and the positive electrode heating drum **16** are different, and are the linear velocity of the positive electrode heating drum **16**>the linear velocity of the positive electrode cutting drum **14**. Accordingly, the electrode cutting head of the positive electrode cutting drum **14** is temporarily accelerated to be roughly the same as the linear velocity of the positive electrode heating drum **16** in front of the position close to the positive electrode heating drum **16**, and turns the relative speed with the positive electrode heating drum **16** to roughly zero. The electrode cutting head of the positive electrode cutting drum **14** discharges the sucked and held positive electrode plate to the side of the positive electrode heating drum **16** at the timing when the relative speed becomes roughly zero. The electrode cutting head of the positive electrode cutting drum **14** is switched to a rotating speed before the acceleration after discharging the sucked and held positive electrode plate. (73) The positive electrode heating drum **16** sucks and holds the positive electrode plate discharged from the positive electrode cutting drum **14**, and heats (preliminarily heats) the positive electrode plate by a built-in heater. The figure illustrates that the positive electrode plate is heated at a position **17**. The heating (preliminary heating) process is for heat-bonding a separator and the positive electrode plate in the subsequent bonding process. The location of heating by the positive electrode heating drum **16** is not limited to a specific position (for example, the position **17**). The positive electrode heating drum **16** may be in the heated state at all times while the drum is rotated. (74) The bonding drum **18** is arranged between the negative electrode heating drum **12** and the positive electrode heating drum **16** so as to be close to both of the negative electrode heating drum **12** and the positive electrode heating drum **16**. The bonding drum **18** has a fifth radius, and is rotated around the central axis at a fifth angular velocity. To the bonding drum **18**, a belt-like separator **S1** is supplied as a first separator single plate, and a belt-like separator **S2** is supplied as a second separator single plate. Further, the negative electrode plate heated by the negative electrode heating drum **12** is supplied and the positive electrode plate heated by the positive electrode heating drum **16** is supplied.

(75) For the negative electrode plate, the linear velocity of the negative electrode heating drum **12** and the linear velocity of the bonding drum **18** are roughly the same, and the heated negative electrode plate sucked and held by the negative electrode heating drum **12** is discharged to the side of the bonding drum **18** at the position close to the bonding drum **18**. In addition, also for the positive electrode plate, the linear velocity of the positive electrode heating drum **16** and the linear velocity of the bonding drum **18** are roughly the same, and the heated positive electrode plate sucked and held by the positive electrode heating drum **16** is discharged to the side of the bonding drum **18** at the position close to the bonding drum **18**.

(76) Note that, since the linear velocity of the negative electrode heating drum **12** and the linear velocity of the bonding drum **18** are roughly the same, the linear velocity of the electrode cutting head of the negative electrode cutting drum **10** in front of the position close to the negative electrode heating drum **12** is roughly the same as the linear velocity of the bonding drum **18**. In addition, since the linear velocity of the positive electrode heating drum **16** and the linear velocity of the bonding drum **18** are roughly the same, the linear velocity of the electrode cutting head of the positive electrode cutting drum **14** in front of the position close to the positive electrode heating drum **16** is roughly the same as the linear velocity of the bonding drum **18**. Accordingly, the electrode cutting head discharges the sucked and held negative electrode plate or positive electrode plate at the timing when the linear velocity becomes roughly the same with the bonding drum **18**.

(77) By the bonding drum **18**, the belt-like separator **S1** is sucked and held at a predetermined position. Thereafter, at the position close to the negative electrode heating drum **12** positioned on a downstream side of a rotation direction, the heated negative electrode plate discharged from the negative electrode heating drum **12** is arranged on the separator **S1**. Then, the belt-like separator **S2** is arranged on the negative electrode plate at a predetermined position on the further downstream

side of the rotation direction. Thereafter, the separator **S1**, the negative electrode plate and the separator **S2** are bonded by being pressurized by a thermocompression bonding roller **19**, the heated positive electrode plate discharged from the positive electrode heating drum **16** is arranged in the separator **S2** at the position near the positive electrode heating drum **16** positioned on the further downstream side of the rotation direction, and the positive electrode plate is bonded by pressing force of the positive electrode heating drum **16**. On the surfaces of the separator **S1** and the separator **S2**, a heat bonding layer where adhesiveness is not expressed at a room temperature but the adhesiveness is expressed by heating is formed. The heat bonding layer is a thermoplastic layer containing a thermoplastic polymer for example, and bonds the belt-like separator **S1** and the negative electrode plate, the negative electrode plate and the belt-like separator **S2**, and the belt-like separator **S2** and the positive electrode plate by utilizing plastic deformation of the thermoplastic polymer by heating. In such a manner, a 4-layer laminated body of the belt-like separator **S1**/the negative electrode plate/the belt-like separator **S2**/the positive electrode plate is generated at the bonding drum **18**. The 4-layer laminated body is conveyed from the bonding drum **18** to the separator cutting drum **20**.

(78) On the other hand, at an interval of every fixed number of pieces, the positive electrode plate is not supplied from the positive electrode cutting drum **14** and the positive electrode plate is not supplied from the positive electrode heating drum **16** either. Therefore, the positive electrode plate is not supplied to the bonding drum **18** at the interval of every fixed number of pieces, and a 3-layer laminated body of the belt-like separator **S1**/the negative electrode plate/the belt-like separator **S2** is generated at the bonding drum **18**. The 3-layer laminated body is conveyed from the bonding drum **18** to the separator cutting drum **20** similarly to the 4-layer laminated body.

(79) The separator cutting drum **20** has a sixth radius, and is rotated around the central axis at a sixth angular velocity. The separator cutting drum **20** comprises a plurality of separator cutting heads arranged in the circumferential direction of the drum. The separator cutting head comprises an outer peripheral surface for sucking and holding the 4-layer laminated body and the 3-layer laminated body, and cutting means. The cutting means are, for example, a blade moved in the direction roughly orthogonal to the circumferential direction of the outer peripheral surface. The conveyed 4-layer laminated body and 3-layer laminated body are sucked and held on the outer peripheral surface and rotated. The separator cutting heads suck and hold the 4-layer laminated body and the 3-layer laminated body so that they are also referred to as holding heads. A gap is formed in the circumferential direction between the plurality of separator cutting heads, and by the movement of the blade loaded on the separator cutting head in the direction roughly orthogonal to the circumferential direction at the gap, the 4-layer laminated body and the 3-layer laminated body sucked and held on the outer peripheral surface are cut to have a predetermined width (third width) by the blade. Specifically, the belt-like separator **S1** and the belt-like separator **S2** are cut between the adjacent negative electrode plates arranged at a predetermined interval of the 4-layer laminated body of the belt-like separator **S1**/the negative electrode plate/the belt-like separator **S2**/the positive electrode plate, or between the adjacent negative electrode plates arranged at the predetermined interval of the 3-layer laminated body of the belt-like separator **S1**/the negative electrode plate/the belt-like separator **S2**, or between the adjacent negative electrode plates arranged at the predetermined interval of the 4-layer laminated body and the 3-layer laminated body. The figure illustrates that cutting is performed at a position **21**.

(80) The laminating drum **22** has a seventh radius, and is rotated around the central axis at a seventh angular velocity. The linear velocity of the laminating drum is adjusted to be roughly the same as the linear velocity of the separator cutting drum **20**. The laminating drum **22** comprises a plurality of laminating heads arranged in the circumferential direction of the drum. The laminating head comprises an outer peripheral surface for sucking and holding the cut 4-layer laminated body and 3-layer laminated body. The plurality of laminating heads are each rotated around the common central axis of the laminating drum **22**, and the individual laminating head is driven by a cam in the

circumferential direction of the drum independent of the other laminating heads. For example, when the two laminating heads adjacent in the circumferential direction are a laminating head 'a' and a laminating head 'b', the laminating head 'a' and the laminating head 'b' are rotated around the common central axis of the drum at the fixed speed, and the mutual relative speed is changed for every predetermined section on the circumference of the laminating drum **22**. For example, the laminating head 'a' and the laminating head 'b' are both rotated at the fixed speed so that the relative speed is 0 at a certain timing, but the laminating head 'a' is accelerated in the direction of separating from the following laminating head 'b' and the relative speed becomes finite at a different timing. By such independent drive of the laminating head, a stop state at a laminating position of a specific laminating head is made possible while maintaining rotation at the fixed angular velocity as the entire laminating drum **22**, and the cut 4-layer laminated body and 3-layer laminated body sucked and held on the outer peripheral surface can be discharged and arranged on a laminating stage **24** in the stop state.

(81) The laminating stage **24** is arranged right under the laminating drum **22**. On the laminating stage **24**, the cut 4-layer laminated body and 3-layer laminated body discharged from the laminating drum **22** are successively laminated and a laminated electrode assembly is formed. The laminating stage **24** is drivable in two axial (X axis and Y axis) direction orthogonal to each other, an inclination angle (θ) on an X-Y plane is adjustable, and thus positioning is performed by adjusting the position (XY position) and the inclination angle (θ) of the cut 4-layer laminated body and 3-layer laminated body discharged from the laminating drum **22**, the cut 4-layer laminated body and 3-layer laminated body are successively laminated, and the laminated electrode assembly is manufactured. The laminating stage **24** comprises claws at four corners, and the laminated 4-layer laminated body and 3-layer laminated body are pressed and fixed by the claws. The laminated 3-layer laminated body and 4-layer laminated body are pressurized and/or heated, and bonded to each other.

(82) An outline of a manufacturing process of the laminated electrode assembly is put in order as follows.

(83) (1) The negative electrode plate is generated by cutting negative electrode single plate on the drum.

(84) (2) The negative electrode plate is heated on the drum.

(85) (3) The positive electrode plate is generated by cutting the positive electrode single plate on the drum.

(86) (4) The positive electrode plate is heated on the drum.

(87) (5) The belt-like separator and the negative electrode plate are bonded on the drum, the belt-like separator is bonded further on the drum, and the positive electrode plate is bonded further on the drum.

(88) (6) The 3-layer laminated body and the 4-layer laminated body are generated by cutting the belt-like separator on the drum.

(89) (7) The 3-layer laminated body and the 4-layer laminated body are laminated by the drum.

(90) (8) The laminated 3-layer laminated body and 4-layer laminated body are pressurized and/or heated to be bonded to each other.

(91) By executing the individual processes of cutting, heating, bonding and laminating on the drums in such a manner, high-speed and continuous processing is made possible.

(92) FIG. 2 illustrates a specific configuration perspective view of the manufacturing device in the embodiment.

(93) In the order from left in the figure, the negative electrode cutting drum **10**, the negative electrode heating drum **12**, the bonding drum **18**, the positive electrode heating drum **16** and the positive electrode cutting drum **14** are closely arranged. The negative electrode cutting drum **10** and the negative electrode heating drum **12** are closely arranged, the negative electrode heating drum **12** and the bonding drum **18** are closely arranged, the positive electrode cutting drum **14** and

the positive electrode heating drum **16** are closely arranged, and the positive electrode heating drum **16** and the bonding drum **18** are closely arranged. The negative electrode cutting drum **10**, the negative electrode heating drum **12**, the bonding drum **18**, the positive electrode heating drum **16** and the positive electrode cutting drum **14** are each rotated around the central axis, and their central axes are approximately parallel to each other.

(94) To the negative electrode cutting drum **10**, the belt-like negative electrode single plate N the tension of which is adjusted by a tension roller is supplied. The negative electrode cutting drum **10** comprises the plurality of cutting heads rotated around the central axis, 12 electrode cutting heads arranged in the circumferential direction for example, and the belt-like negative electrode single plate N is sucked and held on the outer peripheral surface of the electrode cutting heads and rotated together with the electrode cutting heads.

(95) The plurality of electrode cutting heads are each provided with the blade moved in the direction roughly orthogonal to the circumferential direction, that is a width direction of the negative electrode cutting drum, and the blade is moved in the width direction of the drum in a fixed range of the rotation direction and cuts the negative electrode single plate N. After cutting the negative electrode single plate N, the electrode cutting head is moved in the circumferential direction independent of the other electrode cutting heads, the gap between the negative electrode plates is adjusted, and the relative speed is turned to roughly 0 at the position close to the negative electrode heating drum **12**.

(96) The negative electrode plate sucked and held by the electrode cutting head of the negative electrode cutting drum **10** is supplied to the negative electrode heating drum **12** at the position close to the negative electrode heating drum **12**. Specifically, on the outer peripheral surface of the electrode cutting head (holding head), suction holes for sucking and holding the negative electrode plate are formed, and a groove extending in the circumferential direction is formed. To the groove formed on the holding head, supply means to be engaged with the groove to separate the negative electrode plate which is the electrode assembly from a suction surface of the holding head and supply it to the drum of a next stage are arranged between the holding head and the drum of the next stage. As the supply means, for example, a belt conveyor is inserted into the groove at the position close to the negative electrode heating drum **12**, the negative electrode plate sucked and held by the electrode cutting head is made to get on the belt conveyor, and the negative electrode plate is supplied to the side of the negative electrode heating drum **12** via the belt conveyor. The belt conveyor will be further described later. The negative electrode cutting drum **10** rotationally conveys the negative electrode plate at the predetermined interval.

(97) To the negative electrode heating drum **12**, the negative electrode plates cut by the negative electrode cutting drum **10** are successively supplied. The negative electrode heating drum **12** is rotated at the roughly same linear velocity as the bonding drum **18**, heats the negative electrode plates and rotationally conveys the heated negative electrode plates to the position close to the bonding drum **18**.

(98) In addition, to the positive electrode cutting drum **14**, the belt-like positive electrode single plate P the tension of which is adjusted by a plurality of tension rollers is supplied. The positive electrode cutting drum **14** comprises the plurality of cutting heads rotated around the central axis, 12 electrode cutting heads arranged in the circumferential direction for example, and the belt-like positive electrode single plate P is sucked and held on the outer peripheral surface of the electrode cutting heads and rotated together with the electrode cutting heads.

(99) The plurality of electrode cutting heads are each provided with the blade moved in the direction roughly orthogonal to the circumferential direction, that is the width direction of the positive electrode cutting drum, and the blade is moved in the width direction of the drum in the fixed range of the rotation direction and cuts the positive electrode single plate P. After cutting the positive electrode single plate P, the electrode cutting head is moved in the circumferential direction independent of the other electrode cutting heads, the gap between the positive electrode

plates is adjusted, and the relative speed is turned to roughly 0 at the position close to the positive electrode heating drum **16**.

(100) The positive electrode plate sucked and held by the electrode cutting head of the positive electrode cutting drum **14** is supplied to the positive electrode heating drum **16** at the position close to the positive electrode heating drum **16**. Specifically, on the outer peripheral surface of the electrode cutting head (holding head), suction holes for sucking and holding the positive electrode plate are formed, and a groove extending in the circumferential direction is formed. To the groove formed on the holding head, supply means to be engaged with the groove to separate the positive electrode plate which is the electrode assembly from the suction surface of the holding head and supply it to the drum of the next stage are arranged between the holding head and the drum of the next stage. As the supply means, for example, a belt conveyor is inserted into the groove at the position close to the positive electrode heating drum **16**, the positive electrode plate sucked and held by the electrode cutting head is made to get on the belt conveyor, and the positive electrode plate is supplied to the side of the positive electrode heating drum **16** via the belt conveyor.

(101) The positive electrode cutting drum **14** rotationally conveys the positive electrode plate at the predetermined interval.

(102) To the positive electrode heating drum **16**, the positive electrode plates cut by the positive electrode cutting drum **14** are successively supplied. The positive electrode heating drum **16** is rotated at the roughly same linear velocity as the bonding drum **18**, heats the positive electrode plates and rotationally conveys the heated positive electrode plates to the position close to the bonding drum **18**.

(103) To the bonding drum **18**, the belt-like separators **S1** and **S2** the tension of which is adjusted by the plurality of tension rollers are supplied. In addition, the heated negative electrode plates are supplied at the position close to the negative electrode heating drum **12**, and the heated positive electrode plates are supplied at the position close to the positive electrode heating drum **16**. The bonding drum **18** is rotated to be in the same direction as the linear velocity of the negative electrode heating drum **12** and the linear velocity of the positive electrode heating drum **16**. In the order from an upstream side to the downstream side of the rotation direction of the bonding drum **18**, a supply position of the separator **S1**, a supply position of the negative electrode plate, a supply position of the separator **S2** and a supply position of the positive electrode plate are arranged. Between the supply position of the separator **S2** and the supply position of the positive electrode plate, the thermocompression bonding roller **19** is arranged.

(104) The bonding drum **18** sucks and holds the belt-like separator **S1** on the outer peripheral surface and rotationally conveys the separator **S1**. Then, at the position close to the negative electrode heating drum **12**, the heated negative electrode plates are arranged on the separator **S1** at the predetermined interval and pressurized by an inter-drum pressure. The bonding drum **18** holds and rotationally conveys the negative electrode plates arranged at the predetermined interval on the separator **S1**, the belt-like separator **S2** is arranged on the negative electrode plates at the supply position of the separator **S2**, and the 3-layer laminated body of the separator **S1**/the negative electrode plate/the separator **S2** is pressure-bonded by the pressing force by the thermocompression bonding roller **19**.

(105) The 3-layer laminated body thermocompression-bonded by the thermocompression bonding roller **19** is rotationally conveyed further to the position close to the positive electrode heating drum **16**, and the positive electrode plates are arranged at the predetermined interval on the separator **S2** at the position close to the positive electrode heating drum **16** and pressurized by the inter-drum pressure. The positive electrode plates are thermocompression-bonded to the separator **S2** by the inter-drum pressure. In addition, the rotation of the positive electrode cutting drum **14** and the positive electrode heating drum **16** is stopped at the interval of every fixed number of pieces, and the supply of the positive electrode plates from the positive electrode heating drum **16** to the bonding drum **18** is stopped. Thus, the positive electrode plate is not arranged on the separator **S2**

and the 3-layer laminated body is left as it is. The 3-layer laminated body formed of the belt-like separator **S1**/the negative electrode plate/the belt-like separator **S2** and the 4-layer laminated body formed of the belt-like separator **S1**/the negative electrode plate/the belt-like separator **S2**/the positive electrode plate are conveyed to the separator cutting drum **20** via the plurality of tension rollers. Note that, instead of stopping the rotation of the positive electrode cutting drum **14** and the positive electrode heating drum **16** at the timing of obtaining the 3-layer laminated body, the 3-layer laminated body may be created and prepared separately and the 3-layer laminated body may be introduced by a route different from an introducing route of the 4-layer laminated body.

(106) While the 3-layer laminated body and the 4-layer laminated body are thermocompression-bonded by the inter-drum pressure of the negative electrode heating drum **12** and the bonding drum **18**, the inter-drum pressure of the positive electrode heating drum **16** and the bonding drum **18** and the pressing force by the thermocompression bonding roller **19**, when an end of the negative electrode plate cut by the negative electrode cutting drum **10** and an end of the positive electrode plate cut by the positive electrode cutting drum **14** are pressed by the drum or the roller, the end of the negative electrode plate and the end of the positive electrode plate may be damaged. Therefore, by temporarily mitigating pressing by the inter-drum pressure and the thermocompression bonding roller **19** at the end of the negative electrode plate cut by the negative electrode cutting drum **10** and the end of the positive electrode plate cut by the positive electrode cutting drum **14**, damages are prevented. By mitigating the pressing at not only the end of the negative electrode plate cut by the negative electrode cutting drum **10** and the end of the positive electrode plate cut by the positive electrode cutting drum **14** but also the ends on four sides of the negative electrode plate and the ends on four sides of the positive electrode plate, the damages of the negative electrode plate and the positive electrode plate can be suppressed. That is, it is further preferable to bond only the inside of the negative electrode plate and the inside of the positive electrode plate with the separator for suppressing the damages of the negative electrode plate and the positive electrode plate.

(107) The separator cutting drum **20** may be arranged separately from the negative electrode cutting drum **10**, the negative electrode heating drum **12**, the bonding drum **18**, the positive electrode heating drum **16** and the positive electrode cutting drum **14** which are a group of drums.

(108) The separator cutting drum **20** sucks and holds the 3-layer laminated body and the 4-layer laminated body supplied via the tension rollers on the outer peripheral surface, and cuts the belt-like separators **S1** and **S2** of the 3-layer laminated body and the 4-layer laminated body by a plurality of separator cutting head structures similar to the negative electrode cutting drum **10**. The cut 3-layer laminated body and 4-layer laminated body are rotationally conveyed to the position close to the laminating drum **22** while being kept sucked and held by the separator cutting heads. On the outer peripheral surface of the separator cutting head, suction holes for sucking and holding the 3-layer laminated body and the 4-layer laminated body are formed.

(109) The separator cutting drum **20** rotationally conveys the 3-layer laminated body or the 4-layer laminated body at the predetermined interval.

(110) The laminating drum **22** is arranged closely to the separator cutting drum **20**, and is rotated roughly the same as the linear velocity of the separator cutting drum **20**. The laminating drum **22** is configured by the plurality of laminating heads rotated around a drum rotation center. For the laminating head, a longitudinal section shape is roughly T-shaped, suction holes for sucking and holding the 3-layer laminated body and the 4-layer laminated body are formed on the outer peripheral surface, and a vacuum pad is provided inside the suction hole. The 3-layer laminated body and the 4-layer laminated body sucked and held by the separator cutting drum **20** are sucked by the laminating head via the vacuum pad. The plurality of laminating heads are each rotated around the common central axis of the laminating drum **22**, and the individual laminating head is driven in the circumferential direction of the drum independent of the other laminating heads, and is further driven in a radial direction of the drum. That is, the laminating head rotationally conveys

the 3-layer laminated body and the 4-layer laminated body to the position close to the laminating stage **24** while sucking and holding them. When the position close to the laminating stage **24** is reached, the relative speed in the circumferential direction of the drum to the laminating stage **24** becomes 0, and the laminating head is moved in the direction of approaching the laminating stage **24** in the radial direction of the drum. The laminating head brings the sucked and held 3-layer laminated body or 4-layer laminated body into contact with the laminating stage **24** or, in the case where the 3-layer laminated body or the 4-layer laminated body is already laminated on the laminating stage **24**, brings them into contact on the laminated body, turns off sucking and holding force and laminates the sucked and held 3-layer laminated body or 4-layer laminated body. Thereafter, the laminating head is moved in the direction of separating from the laminating stage **24** in the radial direction of the drum again, and restarts the rotation.

(111) The above is the whole description.

(112) Note that, for delivery of the 3-layer laminated body and the 4-layer laminated body from the separator cutting drum **20** to the laminating drum **22**, a method via the vacuum pad provided on the laminating head of the laminating drum **22** has been described but it is not limited thereto. It may be by the method of providing a groove on the separator cutting drum similarly to the positive electrode cutting drum and the negative electrode cutting drum and delivering the 3-layer laminated body and the 4-layer laminated body to the laminating drum **22** via a belt conveyor.

(113) In addition, the positive electrode plates and the negative electrode plates may be delivered from the positive electrode cutting drum and the negative electrode cutting drum to the respective heating drums and the bonding drum not by the belt conveyor but by the vacuum pad. Note that a drum which supplies the electrode assembly or the laminated body of the positive electrode plate and the negative electrode plate or the like to a suction pad is referred to as a first drum, and a drum comprising the suction pad is referred also as a second drum.

(114) Next, details of the manufacturing process of the 3-layer laminated body and the 4-layer laminated body will be described.

(115) FIG. **3** and FIG. **4** schematically illustrate the manufacturing process of the 4-layer laminated body to be a base.

(116) FIG. **3(a)** illustrates a situation of negative electrode cutting at the negative electrode cutting drum **10**. On the negative electrode single plate N comprising the negative electrode current collector and the negative electrode active material layer, tabs Nt are formed at a fixed interval. The tab Nt is formed integrally with the negative electrode current collector, and is formed to project from one edge of the negative electrode current collector (formed to project in the direction orthogonal to an extending direction of the belt-like negative electrode single plate N). The negative electrode cutting drum **10** cuts the negative electrode single plate N at the fixed interval and generates the negative electrode plates NP of the first width. At one edge of the negative electrode plate NP, the tab Nt is formed.

(117) FIG. **3(b)** illustrates the situation of positive electrode cutting at the positive electrode cutting drum **14**. On the positive electrode single plate P comprising the positive electrode current collector and the positive electrode active material layer, tabs Pt are formed at the fixed interval. The tab Pt is formed integrally with the positive electrode current collector, and is formed to project from one edge of the positive electrode current collector (formed to project in the direction orthogonal to the extending direction of the belt-like positive electrode single plate P). The positive electrode cutting drum **14** cuts the positive electrode single plate P at the fixed interval and generates the positive electrode plates PP of the second width. At one edge of the positive electrode plate PP, the tab Pt is formed. A size of the positive electrode plate PP is smaller than the size of the negative electrode plate NP. In addition, the interval (pitch) of the tab Pt of the positive electrode plate PP is smaller than the interval (pitch) of the tab Nt of the negative electrode plate NP.

(118) FIG. **3(c)** illustrates the situation of bonding at the bonding drum **18**. The negative electrode plates NP are arranged at the fixed interval and bonded on the belt-like separator **S1**, the belt-like

separator **S2** is arranged and bonded thereon, and the positive electrode plates **PP** are arranged at the fixed interval and bonded so as to be piled up on the negative electrode plates **NP** further. The positive electrode plate **PP** is arranged inside an existence area of the negative electrode plate **NP**. Even though the negative electrode plates **NP** and the positive electrode plates **PP** are separate from each other, since the belt-like separators **S1** and **S2** are not cut yet and are still in a belt shape, they are the belt-like 4-layer laminated body as a whole.

(119) FIG. 3(d) illustrates the situation of cutting at the separator cutting drum **20**. The separator cutting drum **20** cuts the belt-like 4-layer laminated body at the fixed interval, that is between the adjacent negative electrode plates **NP**, and generates 4-layer laminated bodies **40** of the third width for which the separators **S1** and **S2** are cut off.

(120) FIG. 4 illustrates a configuration of the 4-layer laminated body **40**. The separator **S1** is arranged in a bottom layer, the negative electrode plate **NP** is laminated thereon, the separator **S2** is laminated thereon, and the positive electrode plate **PP** is laminated thereon further.

(121) FIG. 5 and FIG. 6 schematically illustrate the manufacturing process of the 3-layer laminated body.

(122) FIG. 5(a) illustrates the situation of bonding at the bonding drum **18**. The negative electrode plates **NP** are arranged at the fixed interval and bonded on the belt-like separator **S1**, and the belt-like separator **S2** is arranged and bonded thereon. For the positive electrode plate **PP**, the positive electrode plate **PP** is not arranged at the interval of every fixed number of pieces. Even though the negative electrode plates **NP** are separate from each other, since the belt-like separators **S1** and **S2** are not cut yet and are still in the belt shape, they are the belt-like 3-layer laminated body as a whole.

(123) FIG. 5(b) illustrates the situation of cutting at the separator cutting drum **20**. The separator cutting drum **20** cuts the belt-like 3-layer laminated body at the fixed interval, that is between the adjacent negative electrode plates **NP**, and generates 3-layer laminated bodies **30** of the third width for which the separators **S1** and **S2** are cut off. The 3-layer laminated body **30** is generated at the interval of every fixed number of pieces. That is like the 4-layer laminated body **40**, the 4-layer laminated body **40**, the 3-layer laminated body **30**, the 4-layer laminated body

(124) FIG. 6 illustrates the configuration of the 3-layer laminated body **30**. The separator **S1** is arranged in the bottom layer, the negative electrode plate **NP** is laminated thereon, and the separator **S2** is laminated thereon.

(125) As above, the 3-layer laminated body **30** and 4-layer laminated body **40** are generated and are supplied from the separator cutting drum **20** to the laminating drum **22**.

(126) FIG. 7 schematically illustrates a laminating process at the laminating drum **22**.

(127) When the 3-layer laminated body **30** and 4-layer laminated body **40** in a rectangular shape are received from the separator cutting drum **20**, the laminating drum **22** successively arranges and laminates them on the laminating stage **24**. That is, when the 3-layer laminated body **30** of the separator **S1**/the negative electrode plate **NP**/the separator **S2** is received, it is inverted upside down to be the separator **S2**/the negative electrode plate **NP**/the separator **S1** and arranged on the laminating stage **24**.

(128) Next, when the 4-layer laminated body **40** of the separator **S1**/the negative electrode plate **NP**/the separator **S2**/the positive electrode plate **PP** is received, the laminating drum **22** inverts it upside down to be the positive electrode plate **PP**/the separator **S2**/the negative electrode plate **NP**/the separator **S1** and arranges and laminates it on the 3-layer laminated body **30** on the laminating stage **24**. Thus, on the laminating stage **24**, the separator **S2**/the negative electrode plate **NP**/the separator **S1**/the positive electrode plate **PP**/the separator **S2**/the negative electrode plate **NP**/the separator **S1** are laminated. Hereinafter, the laminating drum **22** receives the 4-layer laminated body **40**, inverts it upside down, and arranges and laminates it on the 4-layer laminated body **40** of the laminating stage **24** similarly. Thus, the separator **S2**/the negative electrode plate **NP**/the separator **S1**/the positive electrode plate **PP**/the separator **S2**/the negative electrode plate

NP/the separator S1/the positive electrode plate PP/the separator S2/the negative electrode plate NP/the separator S1/ . . . /the positive electrode plate PP/the separator S2/the negative electrode plate NP/the separator S1 are laminated. The laminating drum **22** manufactures the laminated electrode assembly by successively laminating one 3-layer laminated body **30** and a predetermined number of 4-layer laminated bodies **40** on the laminating stage **24**. By combining and laminating the 3-layer laminated body **30** and the 4-layer laminated bodies **40**, the laminated electrode assembly in the rectangular shape for which electrodes at both ends are always the negative electrode plate NP can be obtained.

(129) Note that the configuration of inversion and lamination is just an example, and it is needless to say that the other lamination methods are also possible. For example, the 4-layer laminated body may be laminated first without being inverted and the 3-layer laminated body may be laminated without being inverted at last.

(130) FIG. **8**-FIG. **10** more specifically illustrate the laminating process of the 3-layer laminated body **30** and the 4-layer laminated body **40**.

(131) FIG. **8** illustrates a separator cutting process at the separator cutting drum **20**. The belt-like 3-layer laminated body **30** and 4-layer laminated body **40** bonded at the bonding drum **18** are cut and separated at a roughly middle position between the adjacent negative electrode plates NP. Similarly, the belt-like 4-layer laminated bodies **40** are also cut and separated from each other at the roughly middle position between the adjacent negative electrode plates NP.

(132) FIG. **9** illustrates the cut 3-layer laminated bodies **30** and 4-layer laminated bodies **40**. When it is assumed that a blank of the positive electrode plate PP is generated at the interval of every 38 pieces of the positive electrode plates PP for example, 37 pieces of the 4-layer laminated bodies **40** follow after one 3-layer laminated body **30**, one 3-layer laminated body **30** is generated again thereafter, and 37 pieces of 4-layer laminated bodies **40** are generated further thereafter. When N is defined as a counter variable, it is the 3-layer laminated body **30** when N=1, it is the 4-layer laminated body **40** when N=2, it is the 4-layer laminated body **40** when N=3, . . . , it is the 4-layer laminated body **40** when N=38, it is the 3-layer laminated body **30** when N=39, and it is the 4-layer laminated body **40** when N=40.

(133) FIG. **10** illustrates the configuration of the laminated electrode assembly formed by arranging the 3-layer laminated body **30** on the laminating stage **24** first and successively laminating the 4-layer laminated bodies **40** thereon. It is the configuration for which one 3-layer laminated body **30** and 37 pieces of the 4-layer laminated bodies **40** are laminated. The electrodes at both ends of the laminated electrode assembly are the negative electrode plate NP.

(134) Next, a specific configuration of the electrode cutting head will be described more in detail.

(135) FIG. **11** illustrates a configuration perspective view of the electrode cutting head configuring the positive electrode cutting drum **14**. The plurality of electrode cutting heads are provided in the circumferential direction centering on a rotating shaft of the positive electrode cutting drum **14**. The electrode cutting head is driven independent of the other electrode cutting heads in the circumferential direction ('a' direction in the figure) by a motor **14v** provided for each electrode cutting head. Here, instead of the motor **14v**, independent drive may be performed using a linear motor, a planet gear, a cogged belt or the like.

(136) On an outer peripheral surface **14p** of the electrode cutting head, suction holes **14g** for sucking and holding the positive electrode single plate P and the positive electrode plate PP (after being cut) are formed. At the almost center in the direction ('b' direction in the figure) orthogonal to the circumferential direction of the outer peripheral surface **14p**, a groove **14r** is formed along the circumferential direction, and the suction holes **14g** are not formed at the groove **14r**. In addition, the electrode cutting head is provided with a cutting mechanism block **14t** comprising round blades **50a** and **50b**. The round blades **50a** and **50b** are a pair of blades. The round blades **50a** and **50b** are a set of upper and lower rotary blades, and cut the positive electrode single plate which is the rectangular electrode assembly by reciprocating in the 'b' direction in the figure while

being rotated. That is, the round blades **50a** and **50b** cut the positive electrode single plate by moving forward in the 'b' direction in the figure from an initial position withdrawn from the outer peripheral surface **14p**, and return to the initial position by moving back in the 'b' direction in the figure thereafter. The cutting mechanism block **14t** comprises a cam **14u** engaged with a cam groove formed on a fixed shaft of the positive electrode cutting drum **14**. The cutting mechanism block **14t** makes the round blades **50a** and **50b** reciprocate along a rail in the direction ('b' direction in the figure) orthogonal to the circumferential direction at a clearance from the adjacent electrode cutting head via a rack and pinion mechanism, by the cam **14u** moving along the cam groove accompanying the rotation of the electrode cutting head. A reciprocating speed in the 'b' direction in the figure of the round blades **50a** and **50b** can be appropriately adjusted by setting a gear ratio of the rack and pinion mechanism.

(137) Further, the cutting mechanism block **14t** is connected to a cam **14w**, and after the round blades **50a** and **50b** are moved forward in the 'b' direction in the figure and cut the positive electrode single plate, the cam **14w** is rotated after the cutting head sucking and holding the positive electrode plate is moved in the direction of separating from the cutting head holding the positive electrode single plate, and thus a rotating shaft **14x** of the cutting mechanism block **14t** is rotated. Then, the cutting mechanism block **14t** and the round blades **50a** and **50b** are moved in the direction of separating from a cut surface of the positive electrode single plate accompanying the rotation of the rotating shaft **14x**. Thereafter, the round blades **50a** and **50b** are moved back in the 'b' direction in the figure.

(138) FIG. **12** illustrates the case of performing cutting by arranging the upper and lower round blades **50a'** and **50b'** so as to overlap with each other in a thickness direction of the positive electrode single plate P. The positive electrode single plate P is configured from a positive electrode current collector **P1** and positive electrode active material layers **P2**, and the round blade **50a'** on an upper side is arranged to such a depth of passing through the positive electrode active material layer **P2** on the upper side and the positive electrode current collector **P1**. In addition, the round blade **50b'** on a lower side is arranged to such a depth of passing through the positive electrode active material layer **P2** on the lower side and the positive electrode current collector **P1**. The depth of the round blade **50a'** and the depth of the round blade **50b'** overlap with each other in the thickness direction of the positive electrode single plate P. The positive electrode single plate P can be cut even by such arrangement positions of the round blades **50a'** and **50b'**, however, the present inventors have confirmed that, by the arrangement of the round blades **50a'** and **50b'**, there are cases where unneeded projections, that are burrs, are generated on the cut surface of the positive electrode single plate P, and degradation of the laminated electrode assembly may be caused.

(139) On the other hand, FIG. **13** illustrates the case of performing cutting by arranging the upper and lower round blades **50a''** and **50b''** so as not to overlap in the thickness direction of the positive electrode single plate P. The round blade **50a''** on the upper side is arranged to such a depth of passing through the positive electrode active material layer **P2** on the upper side but not passing through the positive electrode current collector **P1**. In addition, the round blade **50b''** on the lower side is arranged to such a depth of passing through the positive electrode active material layer **P2** on the lower side but not passing through the positive electrode current collector **P1** similarly, and the upper and lower round blades **50a''** and **50b''** are arranged at a finite distance from each other in the thickness direction. In the case of such an arrangement, the upper and lower positive electrode active material layers **P2** of the positive electrode single plate P are cut by the round blades **50a''** and **50b''**, but the positive electrode current collector **P1** is not cut and is maintained in the belt shape as it is even though a cutout portion is generated at a part of it. However, the positive electrode single plate P is sucked and held on the outer peripheral surfaces of the plurality of electrode cutting heads, and the electrode cutting heads are moved in the circumferential direction independent of each other, adjust the interval between the positive electrode plates PP, and are moved so as to turn the relative linear velocity with the positive electrode heating drum **16** to

roughly 0. When the electrode cutting heads are independently moved in the circumferential direction, mutually reverse tensile force acts on a cutting part of the positive electrode single plate P. By the tensile force, the positive electrode current collector P1 is broken with the cutout portion as an origin and results in being cut. The present inventors have confirmed that, by the arrangement of the round blades 50a'' and 50b'' illustrated in FIG. 13, burrs are not generated or hardly generated on the cut surface of the positive electrode single plate P.

(140) Note that, while the upper and lower round blades 50a'' and 50b'' are arranged so as not to overlap in the thickness direction of the positive electrode single plate P in FIG. 13, the arrangement may be such that the distance in the thickness direction between the upper and lower round blades 50a'' and 50b'' is 0. In other words, when the distance in the thickness direction of the positive electrode single plate P between a blade tip of the upper round blade 50a'' and the blade tip of the lower round blade 50b'' is L and the thickness of the positive electrode current collector P1 is d, the arrangement is performed so as to be $d > L \geq 0$. Here, $L < 0$ means that the upper and lower round blades 50a'' and 50b'' overlap with each other, and means that the upper and lower round blades 50a'' and 50b'' cut the positive electrode active material layers P2 on the upper side and the lower side and the positive electrode current collector P1. It can be said that, when it is $d > L \geq 0$, the positive electrode active material layer P2 on the upper side is cut by the round blade 50a'' on the upper side, the positive electrode active material layer P2 on the lower side is cut by the round blade 50b'' on the lower side, and the positive electrode current collector P1 is broken by the tensile force without being cut.

(141) FIG. 14 illustrates the situation of cutting the positive electrode single plate P at an electrode cutting head 14a and an electrode cutting head 14b relating to FIG. 13.

(142) As illustrated in FIG. 14(a), the electrode cutting head 14a and the electrode cutting head 14b are rotated around the rotating shaft center of the drum while sucking and holding the positive electrode single plate P on the outer peripheral surface. The positive electrode single plate P is configured from the positive electrode current collector P1 and the positive electrode active material layers P2 formed on the both surfaces. When rotational movement is made to a predetermined position, the round blades 50a and 50b are moved forward by the cutting mechanism 14t provided in the electrode cutting head 14a, and the positive electrode active material layers P2 of the positive electrode single plate P are cut.

(143) FIG. 14(b) illustrates the situation where the positive electrode active material layers P2 are cut by the round blades 50a and 50b. The round blades 50a and 50b may form a cut Pc between a first part Pa and a second part Pb of the positive electrode single plate P, wherein the first part Pa and the second part Pb are located on the electrode cutting head 14a and the electrode cutting head 14b adjacent to each other. At the time, the positive electrode current collector P1 is not cut yet, and is kept sucked and held on the outer peripheral surface of the electrode cutting head 14a and the electrode cutting head 14b in the belt shape as it is.

(144) Thereafter, as illustrated in FIG. 14(c), the electrode cutting head 14a is accelerated and moved in the circumferential direction of the drum independent of the electrode cutting head 14b. By the movement, the tensile force is applied to the positive electrode current collector P1, and the belt-like positive electrode current collector P1 is cut. Thereafter, by the cutting mechanism 14t provided in the electrode cutting head 14a, the round blades 50a and 50b are moved back. In this sense, in the present embodiment, it can be said that cutting of the positive electrode single plate P is executed by the combination of the movement of the round blades 50a and 50b and the movement of the electrode cutting heads. Note that details of reciprocation of the round blades 50a and 50b will be further described later using FIG. 15.

(145) In the present embodiment, since the positive electrode current collector P1 of the positive electrode single plate P is cut by the tensile force and the positive electrode active material layers P2 are cut by the round blades 50a and 50b, the cut surface of the positive electrode current collector P1 and the cut surfaces of the positive electrode active material layers P2 are in cut forms

different from each other.

(146) FIG. 16 illustrates a top view and a side view of the positive electrode plate PP after being cut. FIG. 16(a) is the top view, the positive electrode active material layer P2 is present on a surface, and the tab Pt extends from the end. FIG. 16(b) is the side view, and the positive electrode active material layers P2 are present at an upper part and a lower part of the positive electrode current collector P1 similarly to FIG. 14. Here, the cut surface cut by the electrode cutting head is illustrated as c in the figure.

(147) FIG. 17 illustrates a cross section of the cut surface c in FIG. 16(b). FIG. 17 illustrates the cross section of the cut surface c of the positive electrode single plate P cut by the electrode cutting head 14a and the electrode cutting head 14b relating to FIG. 13. FIG. 17 illustrates a range 500a where the round blade 50a is brought into contact by forward movement and a range 500b where the round blade 50b is brought into contact by the forward movement. Since the range where the round blades 50a and 50b rub the positive electrode current collector P1 is relatively small, burrs for which an end face of the positive electrode current collector P1 stretches when the positive electrode current collector P1 is rubbed by the round blades 50a and 50b are suppressed. In addition, as to be described later, by separating the round blades 50a and 50b from the cut surface after cutting, the round blades 50a and 50b do not rub the cut surface when moving back, and damages of the cut surface are suppressed also in this respect.

(148) On the other hand, FIG. 18 schematically illustrates the cut surface c in the case of performing cutting by arranging the round blades 50a and 50b so as to overlap with each other in the thickness direction of the positive electrode single plate P as illustrated in FIG. 12, for comparison. The reciprocation range 500a of the round blade 50a and the reciprocation range 500b of the round blade 50b overlap with each other, and by the round blades 50a and 50b rubbing the positive electrode current collector P1, the end face of the positive electrode current collector P1 stretches and burrs are generated. By comparing FIG. 17 and FIG. 18, an effect of the arrangement of the round blades in the present embodiment is clarified.

(149) The round blades 50a and 50b cut the positive electrode single plate P by reciprocating (i.e., making a forward movement and a return movement sequentially) as already described, and FIG. 15 schematically illustrates the situation of the reciprocation of the round blades 50a and 50b. The round blades 50a and 50b are moved forward in the direction orthogonal to the circumferential direction of the drum from the initial position by the cutting mechanism block 14t and cut the positive electrode active material layer P2. The round blades 50a and 50b are moved forward along a moving plane 52.

(150) Next, the round blades 50a and 50b return to the initial position by return movement. First, after the electrode cutting head 14a sucking and holding the positive electrode plate PP is moved in the direction of separating from the electrode cutting head 14b holding the positive electrode single plate P, the rotating shaft 14x is rotated by the rotation of the cam 14w illustrated in FIG. 11, and the cutting mechanism block 14t and the round blades 50a and 50b are moved in the direction of separating from the positive electrode single plate P accompanying the rotation of the rotating shaft 14x. Thereafter, the round blades 50a and 50b are moved back in the direction orthogonal to the circumferential direction of the drum. After the forward movement, the cutting mechanism block 14t and the round blades 50a and 50b are moved again to the initial position by reverse rotation of the cam 14w. A forward movement track and a return movement track of the round blades 50a and 50b are different from each other, and since the round blades 50a and 50b are not brought into contact with the cut surface of the positive electrode single plate P and the positive electrode plate PP during the return movement, damages of the cut surface can be effectively suppressed.

(151) In the present embodiment, the round blades 50a and 50b are illustrated as the blades of the electrode cutting head, however, the shape of the blades is not limited thereto. In addition, in the present embodiment, damages of the cut surface are suppressed by making the tracks of the reciprocation of the round blades 50a and 50b of the electrode cutting head be different, however,

the configuration of suppressing damages of the cut surface is not limited thereto.

(152) FIG. **19** illustrates another blade shape of the electrode cutting head. The upper and lower blades **50a'''** and **50b'''** are round blades as a whole, however, a part of the blade is a flat portion. When paying attention to the blade **50a'''**, the blade **50a'''** is configured from an arc portion **50a1** and a flat portion **50a2**, the blade is formed at the arc portion **50a1**, and the blade is not formed at the flat portion **50a2**. It is similar for the blade **50b'''**, the blade **50b'''** is configured from an arc portion **50b1** and a flat portion **50b2**, the blade is formed at the arc portion **50b1**, and the blade is not formed at the flat portion **50b2**. The blades **50a'''** and **50b'''** are both rotated centering on the rotating shaft, and cut the positive electrode single plate P at the arc portions **50a1** and **50b1** where the blade is formed.

(153) FIG. **20** illustrates the situation of cutting the positive electrode single plate P by the rotation of the blades **50a'''** and **50b'''** illustrated in FIG. **19**. The arc portion **50a1** of the blade **50a'''** and the arc portion **50b1** of the blade **50b'''** face each other at the initial position, and the positive electrode single plate P is cut by the arc portion **50a1** and the arc portion **50b1** as illustrated in (a) and (b) by the rotation of the blades **50a'''** and **50b'''** from the initial position during the forward movement. The blades **50a'''** and **50b'''** are further rotated, and when they are rotated to the position where the flat portion **50a2** and the flat portion **50b2** face each other as in (c), the cutting of the positive electrode single plate P is ended and the forward movement is completed. During the return movement, while maintaining the state where the flat portion **50a2** and the flat portion **50b2** face each other, the blades **50a'''** and **50b'''** are moved to the initial position. Since an inter-axial distance of the blades **50a'''** and **50b'''** is fixed, in the state where the flat portion **50a2** and the flat portion **50b2** face each other, a gap is generated between the blades **50a'''** and **50b'''**. By being moved back while maintaining the gap, the blades **50a'''** and **50b'''** are not brought into contact with the cut surface and damages of the cut surface can be suppressed. In this case, the need of the movement (inclination) of the cutting mechanism block **14t** by the rotation of the cam **14w** as illustrated in FIG. **15** can be eliminated.

(154) The positive electrode plate PP is generated by cutting the positive electrode single plate P at the positive electrode cutting drum **14** as above, and the generated positive electrode single plate PP is supplied to the positive electrode heating drum **16**. The positive electrode plate PP can be supplied from the positive electrode cutting drum **14** to the positive electrode heating drum **16** via a belt conveyor for example.

(155) FIG. **21** schematically illustrates supply of the positive electrode plate PP from the positive electrode cutting drum **14** to the positive electrode heating drum **16**. In addition, FIG. **22** illustrates a partially enlarged view of FIG. **21**, that is an enlarged view of a contact part of the positive electrode cutting drum **14** and the positive electrode heating drum **16**.

(156) A belt conveyor **140** is arranged near the contact part of the positive electrode cutting drum **14** and the positive electrode heating drum **16**, and one end of the belt conveyor **140** is inserted into the groove **14r** of the positive electrode cutting drum **14** (see FIG. **11**). A width of a pulley on one end side of the belt conveyor **140** inserted into the groove **14r** is roughly the same as the width of the groove **14r**. A pulley on the other end side of the belt conveyor **140** is arranged near the positive electrode heating drum **16**. A belt of the belt conveyor **140** extends from the groove **14r** of the positive electrode cutting drum **14** to vicinity of the positive electrode heating drum **16**, turns back at the pulley on the other end side of the belt conveyor **140**, and returns into the groove **14r** of the positive electrode cutting drum **14**.

(157) The positive electrode single plate PP is cut at the positive electrode cutting drum **14**, sucked and held on the outer peripheral surface, and rotationally conveyed. When the positive electrode plate PP is rotationally conveyed and brought into contact with the belt conveyor **140** one end of which is inserted into the groove **14r**, the positive electrode plate PP gets on the belt conveyor **140**, separates from the surface of the positive electrode cutting drum **14** and rides on the belt conveyor **140**. The positive electrode plate PP riding on the belt conveyor **140** is conveyed to the positive

electrode heating drum **16** by the belt conveyor **140**, sucked by the suction holes formed on the outer peripheral surface of the positive electrode heating drum **16**, moved from the belt conveyor **140** to the outer peripheral surface of the positive electrode heating drum **16**, sucked and held. Note that, while the belt conveyor **140** inserted into the groove **14r** is held by the pulley on one end side in the description above, a knife edge may be inserted into the groove **14r** instead of the pulley and the belt conveyor **140** inserted into the groove **14r** may be held by the knife edge.

(158) The cutting of the positive electrode single plate P at the positive electrode cutting drum **14** and the supply of the positive electrode plate PP from the positive electrode cutting drum **14** to the positive electrode heating drum **16** have been described above, and it is similar for the cutting of the negative electrode single plate N at the negative electrode cutting drum **10** and the supply of the negative electrode plate NP from the negative electrode cutting drum **10** to the negative electrode heating drum **12**. In addition, also for the cutting of the belt-like separator at the separator cutting drum **20**, by making the blades **50a** and **50b** move back without being in contact with the cut surface, damages of the cut surface can be suppressed.

(159) Next, the process of supplying the negative electrode plate NP or the positive electrode plate PP from the negative electrode heating drum **12** or the positive electrode heating drum **16** to the bonding drum **18** will be described in detail.

(160) FIG. **23A** illustrates the negative electrode heating drum **12** and the bonding drum **18**. The negative electrode heating drum **12** receives the negative electrode plate NP cut at the negative electrode cutting drum **10**, sucks and holds it on the outer peripheral surface, and heats it while rotationally conveying it. Then, at the position close to the bonding drum **18**, the heated negative electrode plate NP is successively bonded at the fixed interval on the belt-like separator S1 which is sucked and held on the outer peripheral surface of the bonding drum **18** and rotationally conveyed.

(161) The negative electrode heating drum **12** is pivotally supported freely swingably centering on a drum rotation fulcrum **122**, and is swung centering on the drum rotation fulcrum **122** by a motor **124**. In addition, to the negative electrode heating drum **12**, a coil spring **120** is installed so as to face the drum rotation fulcrum **122**. When the negative electrode heating drum **12** is swung centering on the rotation fulcrum **122**, a deflection amount of the coil spring **120** changes, and thus pressurizing force at a pressurizing point which is a contact point of the negative electrode heating drum **12** and the bonding drum **18** is adjusted.

(162) While the negative electrode heating drum **12** presses the heated negative electrode plate NP to the belt-like separator S1 on the bonding drum **18** and bonds the negative electrode plate NP to the separator S1 by the pressurizing force at the pressurizing point, a finite eccentricity exists at the negative electrode heating drum **12** and the finite eccentricity also exists at the bonding drum **18**. When the eccentricity exists, since the deflection amount of the coil spring **120** changes, the negative electrode plate NP cannot be pressurized and bonded by fixed pressurizing force.

(163) Then, eccentricity amounts of the negative electrode heating drum **12** and the bonding drum **18** are combined, the negative electrode heating drum **12** is swung centering on the rotation fulcrum **122** by rotating the motor **124** at all times using the combined value matched with drum rotation, the change of the pressurizing force at the pressurizing point is offset, and the negative electrode plate NP is pressurized and bonded on the belt-like separator S1 by almost fixed pressurizing force.

(164) The eccentricity amounts of the negative electrode heating drum **12** and the bonding drum **18** are measured and obtained beforehand, and stored in a memory of a controller as a table. Drive signals based on a cancellation waveform which cancels a composite waveform of the eccentricity amounts of the two drums are supplied to the motor **124**, deflection by the eccentricity of the coil spring **120** is suppressed and fluctuation of the pressurizing force is suppressed.

(165) By driving the motor **124** with the cancellation waveform as a drive signal waveform, the controller can cancel the respective eccentricities and press the negative electrode heating drum **12** to the bonding drum **18** by the fixed pressurizing force at all times.

(166) However, if the negative electrode heating drum **12** is pressed to the bonding drum **18** by the fixed pressurizing force and the negative electrode plate NP is bonded onto the belt-like separator **S1**, there is a risk that the negative electrode plate NP is damaged by the pressurizing force at the end of the negative electrode plate NP, that is the end in the circumferential direction of the drum, in particular. Therefore, damages of both ends of the negative electrode plate NP are prevented by reducing the pressurizing force between the negative electrode heating drum **12** and the bonding drum **18** or interrupting the pressurization (turning the pressurizing force to 0) at both ends in the circumferential direction of the negative electrode plate NP without bonding the negative electrode plate NP onto the separator **S1** by the fixed pressurizing force.

(167) FIG. **24** illustrates the situation of bonding the negative electrode plate NP onto the belt-like separator **S1** by the fixed pressurizing force. That is, FIG. **24** illustrates the situation of bonding the negative electrode plate NP onto the separator **S1** without reducing the pressurizing force between the negative electrode heating drum **12** and the bonding drum **18** or interrupting the pressurization. The separator **S1** is linearly illustrated in FIG. **24**, but is actually sucked and held by the bonding drum **18** and is in the state of drawing an arc along an arc-shaped outer shape of the bonding drum **18**. In addition, one negative electrode plate NP is heated on the negative electrode heating drum **12** in FIG. **24**, however, the plurality of negative electrode plates NP can be heated.

(168) In FIG. **24(a)**, the negative electrode heating drum **12** is in direct contact with the separator **S1** in an area where the negative electrode plate NP is not present, and presses the separator **S1** by the fixed pressurizing force. When the negative electrode plate NP is rotationally conveyed in the state, as illustrated in FIG. **24(b)**, the negative electrode heating drum **12** gets on a circumferential direction tip end of the negative electrode plate NP first and presses the circumferential direction tip end to the separator **S1** by the fixed pressurizing force. At the time, the tip end of the negative electrode plate NP is crushed and deformed by the pressurizing force.

(169) Even thereafter, the negative electrode heating drum **12** continuously presses the negative electrode plate NP to the separator **S1** by the fixed pressurizing force, and bonds the negative electrode plate NP to the separator **S1**. Then, as illustrated in FIG. **24(c)**, a circumferential direction rear end of the negative electrode plate NP is also pressed to the separator **S1** by the fixed pressurizing force so that the rear end is also crushed and deformed by the pressurizing force similarly to the tip end.

(170) In such a manner, when the negative electrode heating drum **12** is pressed to the bonding drum **18** by the fixed pressurizing force at all times, there is a risk that the separator **S1** is damaged since the separator **S1** is pressed even at a part where the negative electrode plate NP is not present, and also there is a risk that the circumferential direction tip end and the circumferential direction rear end of the negative electrode plate NP are pressed and damaged. FIG. **24(c)** schematically illustrates crushing at the circumferential direction tip end and the circumferential direction rear end as a slope shape.

(171) On the other hand, FIG. **25** illustrates the situation of bonding the negative electrode plate NP to the separator **S1** in the present embodiment. That is, FIG. **25** illustrates the situation of bonding the negative electrode plate NP onto the separator **S1** with the timing of reducing the pressurizing force between the negative electrode heating drum **12** and the bonding drum **18** or interrupting the pressurization. The separator **S1** is linearly illustrated in FIG. **25**, but is actually sucked and held by the bonding drum **18** and is in the state of drawing an arc along the arc-shaped outer shape of the bonding drum **18**. In addition, one negative electrode plate NP is heated on the negative electrode heating drum **12** also in FIG. **25**, however, the plurality of negative electrode plates NP can be heated.

(172) In FIG. **25(a)**, the negative electrode heating drum **12** regulates the pressurizing force at the pressurizing point in the area where the negative electrode plate NP is not present so as not to press the negative electrode heating drum **12** to the bonding drum **18**. Even when the negative electrode plate NP is rotationally conveyed and the circumferential direction tip end of the negative electrode

plate NP is brought into contact with the separator S1 in the state, regulation of the fixed pressurizing force is maintained as it is. Accordingly, the circumferential direction tip end of the negative electrode plate NP is not pressed.

(173) In FIG. 25(b), when the negative electrode plate NP is rotationally conveyed further and a center portion other than the circumferential direction tip end is brought into contact with the separator S1, the negative electrode heating drum 12 is pressed to the bonding drum 18 by the fixed pressurizing force. Thus, the negative electrode plate NP is bonded to the separator S1.

(174) In FIG. 25(c), when the negative electrode plate NP is rotationally conveyed further and the circumferential direction rear end is brought into contact with the separator S1, the pressurizing force is regulated and the negative electrode heating drum 12 is not pressed to the bonding drum 18. Accordingly, the circumferential direction rear end of the negative electrode plate NP is not pressed.

(175) As above, by pressing only the center portion of the negative electrode plate NP by the fixed pressurizing force, regulating the fixed pressurizing force at both ends, that are the circumferential direction tip end and the circumferential direction rear end, of the negative electrode plate NP and not pressing both ends, damages of both ends of the negative electrode plate NP are prevented. In addition, by regulating the fixed pressurizing force also in the area where the negative electrode plate NP is not present, damages of the separator S1 are also prevented.

(176) Separately from cancellation of the eccentricity amounts of the negative electrode heating drum 12 and the bonding drum 18, the controller reduces the pressurizing force between the negative electrode heating drum 12 and the bonding drum 18 or interrupts the pressurization at both ends of the negative electrode plate NP. A mechanism of reducing the pressurizing force between the negative electrode heating drum 12 and the bonding drum 18 or interrupting the pressurization will be described based on FIG. 23B and FIG. 23C.

(177) FIG. 23B and FIG. 23C are a perspective view and a rear elevation of a pressurizing force adjusting mechanism, and a column α of the bonding drum 18 is provided with a cam mechanism 80. To a column β of the negative electrode heating drum 12, a swing arm 82 is attached as an arm engaged with the cam mechanism 80, and the swing arm 82 pivotally supports a roller 84. The surface of the cam mechanism 80 and the roller 84 are in contact. A surface shape of the cam mechanism 80 is polygonal or has projections at a predetermined interval. In the cam mechanism 80, corner portions of a polygon or the projections operate to reduce the pressurizing force between the negative electrode heating drum 12 and the bonding drum 18 or to interrupt the pressurization. The cam mechanism 80 is driven by a motor independent of the drive of the motor 124. By the operation of the cam mechanism 80, a distance between the negative electrode heating drum 12 and the bonding drum 18 is controlled, and the pressurizing force to the negative electrode plate NP can be reduced or the pressurization can be interrupted.

(178) The pressurizing force between the negative electrode heating drum 12 and the bonding drum 18 is kept fixed by the drive of the motor 124, however, the distance between the column β to which the negative electrode heating drum 12 is attached and the column α to which the bonding drum 18 is attached fluctuates. Accompanying the fluctuation of the distance, the fluctuation of the distance between the cam mechanism 80 and the roller 84 is generated and there may be a case where the pressurizing force between the negative electrode heating drum 12 and the bonding drum 18 cannot be reduced or the pressurization cannot be interrupted at a predetermined timing.

(179) In order to keep the distance between the cam mechanism 80 and the roller 84 fixed, the swing arm 82 is provided with a roller 86, and an inclination member 88, a roller 90 and a cam mechanism 92 drivable independent of the column β of the negative electrode heating drum 12 are provided on the side of the column β . The roller 86 is in contact with an inclined surface of the inclination member 88. The roller 90 is attached to the inclination member 88 and is in contact with the cam mechanism 92. Since the inclination member 88 is moved up and down by the cam mechanism 92 driven by the motor, the swing arm 82 is swung via the roller 86 in contact with the

inclined surface of the inclination member **88**.

(180) While the column β and the inclination member **88** are moved to right and left by the drive of the motor **124** according to a cancellation waveform, the inclined surface of the inclination member **88** is also moved up and down by the drive of the motor (not illustrated) which moves the cam mechanism **92** up and down also according to the cancellation waveform, and thus the fluctuation of the position of the roller **86** can be suppressed. Since the fluctuation of the position of the roller **86** is suppressed, the fluctuation of the distance between the roller **84** and the cam mechanism **80** can be suppressed.

(181) Note that the method of bonding the negative electrode plate NP to the belt-like separator **S1** by the fixed pressurizing force by the drive of the motor **124** and suppressing damages of both ends of the negative electrode plate NP by the drive of the cam mechanism **80** has been described in the description above, however, it is possible to suppress damages of both ends of the negative electrode plate NP after bonding the negative electrode plate NP to the belt-like separator **S1** by the fixed pressurizing force even by driving the motor **124**.

(182) The positions of the negative electrode plates NP and the interval between the negative electrode plates NP at the negative electrode heating drum **12** can be detected by a camera arranged near the negative electrode heating drum **12** for example. The controller which monitors and controls the entire manufacturing device can also receive position data and interval data of the negative electrode plate NP detected by the camera, and control permission/regulation of the fixed pressurizing force (or ON/OFF of the fixed pressurizing force) using the received position data and interval data.

(183) It is similar for a relation of the positive electrode heating drum **16** and the bonding drum **18**, and the controller defines the drive signal waveform for cancelling the eccentricity amounts of the positive electrode heating drum **16** and the bonding drum **18** and pressing the positive electrode heating drum **16** to the bonding drum **18** by the fixed pressurizing force as a reference, superimposes the drive signal waveform for regulating the fixed pressurizing force at both ends of the positive electrode plate PP and in the area where the positive electrode plate PP is not present, and performs driving.

(184) Further, while the thermocompression bonding roller **19** is a roller for pressing and thermocompression-bonding the belt-like separator **S1**/the negative electrode plate NP/the belt-like separator **S2**, the controller can also drive the roller by the drive signal waveform for regulating the fixed pressurizing force at both ends of the negative electrode plate NP and in the area where the negative electrode plate NP is not present.

(185) Note that, while the case where the drum rotation fulcrum **122**, the motor **124** and the coil spring **120** are installed to the negative electrode heating drum **12** has been described in the description relating to FIG. **23A**, they may be provided on the side of the bonding drum **18**. However, while FIG. **23A** illustrates the state where the negative electrode heating drum **12** and the bonding drum **18** are in contact, there is a case where the drum which supplies the positive electrode plate is also in contact with the bonding drum **18**. In such a case, the drum rotation fulcrum, the motor and the coil spring are preferably provided on the drum which supplies the positive electrode plate and the drum which supplies the negative electrode plate.

(186) FIG. **26**-FIG. **28** illustrate a pressurizing range at the negative electrode heating drum **12**, the thermocompression bonding roller **19** and the positive electrode heating drum **16**.

(187) FIG. **26** illustrates the pressurizing range when bonding the belt-like separator **S1** and the negative electrode plate NP by the negative electrode heating drum **12** and the bonding drum **18** by a dashed line. Of the negative electrode plate NP, both ends in a longitudinal direction of the belt-like separator **S1** are not pressurized, and the other area is pressurized and pressed. In other words, of four sides of the negative electrode plate NP in the rectangular shape, the two opposite sides, that are the two opposite sides in the longitudinal direction of the belt-like separator **S1**, are not pressurized, and the two opposite sides in the direction orthogonal to the longitudinal direction are

pressurized. Of the four sides of the negative electrode plate NP in the rectangular shape, the two pressurized sides are bonded with the separator S1 at the pressurized parts, and the two sides that are not pressurized are not bonded with the separator S1.

(188) FIG. 27 illustrates the pressurizing range when bonding the belt-like separator S1, the negative electrode plate NP and the belt-like separator S2 by the thermocompression bonding roller 19 and the bonding drum 18 by a dashed line. Similarly to the case of FIG. 26, of the negative electrode plate NP, both ends in the longitudinal direction of the belt-like separators S1 and S2 are not pressurized, and the other area is pressurized and pressed. In other words, of the four sides of the negative electrode plate NP in the rectangular shape, the two opposite sides, that are the two opposite sides in the longitudinal direction of the belt-like separators S1 and S2, are not pressurized, and the two opposite sides in the direction orthogonal to the longitudinal direction are pressurized. Of the four sides of the negative electrode plate NP in the rectangular shape, the two pressurized sides are bonded with the separator S2 at the pressurized parts, and the two sides that are not pressurized are not bonded with the separator S1.

(189) FIG. 28 illustrates the pressurizing range when bonding the belt-like separator S1, the negative electrode plate NP, the belt-like separator S2 and the positive electrode plate PP by the positive electrode heating drum 16 and the bonding drum 18 by a dashed line. Of the positive electrode plate PP, both ends in the longitudinal direction of the belt-like separators S1 and S2 are not pressurized, and the other area is pressurized and pressed. In other words, of the four sides of the positive electrode plate PP in the rectangular shape, the two opposite sides, that are the two opposite sides in the longitudinal direction of the belt-like separators S1 and S2, are not pressurized, and the two opposite sides in the direction orthogonal to the longitudinal direction are pressurized. Of the four sides of the positive electrode plate PP in the rectangular shape, the two pressurized sides are bonded with the separators S1 and S2 at the pressurized parts, and the two sides that are not pressurized are not bonded with the separators S1 and S2.

(190) Since the size of the positive electrode plate PP is smaller than the size of the negative electrode plate NP, the pressurizing range in FIG. 28 is smaller than the pressurizing range in FIG. 26. As above, the 3-layer laminated body 30 and the 4-layer laminated body 40 are manufactured while the pressurizing range is controlled.

(191) Next, the laminating process at the laminating drum 22 and the laminating stage 24 will be described in more detail.

(192) FIG. 29 illustrates a configuration perspective view of the laminating drum 22. The laminating drum 22 is arranged closely to the separator cutting drum 20, and is rotated roughly the same as the linear velocity of the separator cutting drum 20. The laminating drum 22 is configured by the plurality of laminating heads which are rotated around the drum rotation center. The number of the laminating heads is arbitrary, and the laminating drum 22 is configured from 12 laminating heads 22a, 22b, 22c, 22d, 22e, 22f, 22g, 22h, 22i, 22j, 22k and 22m for example. Each laminating head is configured from an arm (spoke) connected to a drum center axis (hub) and a holding portion connected to the other end of the spoke. For each laminating head, the longitudinal section shape is roughly T-shaped by the arm and the holding portion, and the suction holes for sucking and holding the 3-layer laminated body 30 and the 4-layer laminated body 40 generated by being cut by the separator cutting drum 20 are formed on the outer peripheral surface of the holding portion. The holding portion is connected freely swingably in the circumferential direction to the arm.

(193) The plurality of laminating heads 22a-22m are each rotated around the central axis of the laminating drum 22, and the individual laminating head is driven in the circumferential direction of the drum independent of the other laminating heads, and is further driven in the radial direction of the drum. That is, the individual laminating head rotationally conveys the 3-layer laminated body 30 and the 4-layer laminated body 40 to the position close to the laminating stage 24 while sucking and holding them. When the position close to the laminating stage 24 is reached, the relative speed

in the circumferential direction of the drum to the laminating stage **24** becomes 0, and the individual laminating head is moved in the direction of approaching the laminating stage **24** in the radial direction of the drum. The individual laminating head which approaches the laminating stage **24** brings the sucked and held 3-layer laminated body **30** or 4-layer laminated body **40** into contact with the laminating stage **24** or, in the case where the 3-layer laminated body **30** or the 4-layer laminated body **40** is already laminated on the laminating stage **24**, into contact on the laminated body, turns off the sucking and holding force and laminates the sucked and held 3-layer laminated body **30** or 4-layer laminated body **40**. Thereafter, the individual laminating head is moved in the direction of separating from the laminating stage **24** in the radial direction of the drum.

(194) FIG. **30**-FIG. **33** illustrate a basic operation of the laminating heads **22a-22m**. Hereinafter, in the description of the operation of the individual laminating head, “the radial direction of the drum” and “the radial direction of the laminating drum **22**” are appropriately referred to as “the radial direction”.

(195) As illustrated in FIG. **30**, the laminating heads **22a-22m** are rotated around the rotation center axis of the laminating drum **22** at the fixed angular velocity, and when the predetermined position is reached, accelerated in the circumferential direction independent of the other laminating heads. For example, when paying attention to the laminating head **22a**, the laminating head **22a** is accelerated when the predetermined position in the circumferential direction is reached, approaches the adjacent laminating head **22m** on the upstream side of the rotation direction, and separates from the adjacent laminating head **22b** on the downstream side of the rotation direction.

(196) After the acceleration, when the laminating head **22a** reaches the position close to the laminating stage **24** as illustrated in FIG. **31**, for the laminating head **22a**, the relative speed in the circumferential direction of the drum to the laminating stage **24** becomes 0. At the close position, the 3-layer laminated body **30** or the 4-layer laminated body **40** sucked and held on the outer peripheral surface is laminated on the laminating stage **24**. In more detail, when approaching the position close to the laminating stage **24**, the outer peripheral surface of the holding portion is swung to an angle to be roughly parallel to a stage surface of the laminating stage **24**, and the laminating head **22a** brings the outer peripheral surface of the holding portion close to the stage surface of the laminating stage **24** by being moved in the direction of the laminating stage **24** by the arm moving in the radial direction while maintaining the state that the outer peripheral surface of the holding portion is roughly parallel to the stage surface of the laminating stage **24**, and laminates the 3-layer laminated body **30** or the 4-layer laminated body **40** sucked and held on the outer peripheral surface onto the laminating stage **24**.

(197) FIG. **32** and FIG. **33** illustrate the state after the 3-layer laminated body **30** or the 4-layer laminated body **40** is laminated on the laminating stage **24**. The adjacent laminating head **22b** on the rotation direction downstream side of the laminating head **22a** is accelerated since it has reached the predetermined position, and approaches the laminating head **22a**. After laminating the 3-layer laminated body **30** or the 4-layer laminated body **40** on the laminating stage **24**, the laminating head **22a** is rotated in the circumferential direction and withdrawn from the position close to the laminating stage **24**. In more detail, the laminating head **22a** is moved in such a direction that the outer peripheral surface separates from the position close to the stage surface of the laminating stage **24** by the arm moving in the radial direction, and is moved in the rotation direction and withdrawn so that the laminating head **22a** does not interfere with the laminating head **22b**. After withdrawal from the close position, the outer peripheral surface of the holding portion is swung in the opposite direction and returns.

(198) FIG. **34** illustrates a specific configuration example of the laminating head **22a**. It is similar also for the other laminating heads **22b-22m**.

(199) The laminating head **22a** comprises two arms **22a1** and **22a2** and a holding portion **22a3**. The arms **22a1** and **22a2** extend in parallel to the radial direction of the laminating drum **22**, the arm **22a1** and the arm **22a2** are rotationally driven integrally in the circumferential direction of the

drum by a cam mechanism **22a4**, and the arm **22a1** and the arm **22a2** are driven to swing in the circumferential direction by a cam mechanism **22a5**. On ends in the radial direction of the arm **22a1** and the arm **22a2**, an outer peripheral surface **22a3** is freely swingably provided. The outer peripheral surface **22a3** is rotated together with the rotation in the circumferential direction of the arm **22a1** and the arm **22a2**, and is swung/moved together with the swing and radial direction movement of the arm **22a1** and the arm **22a2**. Further, the outer peripheral surface **22a3** is swung to the arm **22a1** and the arm **22a2** by a cam mechanism **22a6**.

(200) FIG. **35** illustrates a time change of the position of the laminating head **22a**. In the figure, a horizontal axis indicates time, and a vertical axis indicates the position (angle) in the circumferential direction of the drum.

(201) From a certain reference position to a position $\theta 1$, the laminating head **22a** is rotated at the fixed angular velocity. The outer peripheral surface **22a3** is rotated while sucking and holding the 3-layer laminated body **30** or the 4-layer laminated body **40**.

(202) Next, when the predetermined position $\theta 1$ is reached, the laminating head **22a** is accelerated in the circumferential direction, and the outer peripheral surface **22a3** is also accelerated accompanying that. During the time, since the adjacent laminating head **22b** on the rotation direction downstream side is continuously rotated at the fixed angular velocity, a clearance between the laminating head **22a** and the laminating head **22b** increases. It can be said that, by the acceleration, the laminating head **22a** generates/secures stop time at the position close to the laminating stage **24**, that is lamination time of the 3-layer laminated body **30** or the 4-layer laminated body **40** onto the laminating stage **24**. In addition, during the acceleration, the outer peripheral surface **22a3** is swung to the arms **22a1** and **22a2**, and the surface sucking and holding the 3-layer laminated body **30** or the 4-layer laminated body **40** is inclined so as to be roughly parallel to the stage surface of the laminating stage **24**.

(203) Next, when a position $\theta 2$ close to the laminating stage **24** is reached, the laminating head **22a** is stopped at the position (needless to say, deceleration control is executed before stop, however, it is omitted for convenience of the description), moves the arm **22a1** and the arm **22a2** in the radial direction to bring the outer peripheral surface close to the stage surface of the laminating stage **24**, turns off suction force of the outer peripheral surface, and laminates the 3-layer laminated body **30** or the 4-layer laminated body **40** on the laminating stage **24**. At the time, by the claws provided on the laminating stage **24**, the corner parts of the 3-layer laminated body **30** or the 4-layer laminated body **40** supplied from the laminating head **22a** are pressed and held. The movement of the claws of the laminating stage **24** linked with the movement of the laminating head **22a** will be further described later.

(204) After laminating the 3-layer laminated body **30** or the 4-layer laminated body **40**, the laminating head **22a** is accelerated and rotated so as to withdraw to a predetermined position $\theta 3$. During the withdrawal, the outer peripheral surface **22a3** is swung to the arms **22a1** and **22a2**, and the surface sucking and holding the 3-layer laminated body **30** or 4-layer laminated body **40** is inclined and returned to an initial angle.

(205) Then, when the predetermined position $\theta 3$ is reached, the laminating head **22a** is rotated at the fixed angular velocity again until the predetermined position $\theta 1$ is reached again. During the period, the laminating head **22a** receives a new 3-layer laminated body **30** or 4-layer laminated body **40** from the separator cutting drum **20**, sucks and holds it on the outer peripheral surface **22a3**, and then reaches the predetermined position $\theta 1$.

(206) Next, the operation of the laminating stage **24** will be described in more detail.

(207) FIG. **36** illustrates a configuration perspective view of the laminating stage **24**. The laminating stage **24** is arranged right under the laminating drum **22**. The laminating stage **24** comprises a stage surface **24g** where the 3-layer laminated body **30** and the 4-layer laminated body **40** are laminated, and claws **24a-24d** which press the 3-layer laminated body **30** and the 4-layer laminated body **40** laminated on the stage surface **24g** from above and hold them.

(208) The stage surface **24g** forms a rectangular outer shape corresponding to the rectangular shape of the 3-layer laminated body **30** and the 4-layer laminated body **40**, and is arranged within a roughly horizontal plane. The 3-layer laminated body **30** or the 4-layer laminated body **40** sucked and held on the outer peripheral surface of the laminating heads **22a-22m** of the laminating drum **22** is positioned on the rectangular stage surface **24g** and laminated. Specifically, when two orthogonal axes within the horizontal plane are defined as an X axis and a Y axis and the rotation direction of the X axis with a certain direction within the horizontal plane as a reference is defined as a 0 direction, the stage surface **24g** is driven in X axis and Y axis directions and also driven in the 0 direction, that is, moved and rotationally driven within the horizontal plane, and is thus positioned so as to match a sucked and held posture of the 3-layer laminated body **30** or the 4-layer laminated body **40** sucked and held on the outer peripheral surface of the laminating heads **22a-22m** of the laminating drum **22**. For a lamination order, for example, the 3-layer laminated body **30** is arranged on the stage surface **24g** first, then the 4-layer laminated body **40** is laminated on the 3-layer laminated body **30** and then the new 4-layer laminated body **40** is laminated on the already laminated 4-layer laminated body **40**.

(209) The claws **24a-24d** are arranged respectively at four corners of the rectangular stage surface **24g**. The claws **24a-24d** each form a roughly L-shaped planar shape, are supported freely swingably in the θ direction centering on the shaft, and are also supported freely movably in a vertical direction (hereinafter, it is appropriately referred to as a vertically upper direction or a vertically lower direction). A drive source of the claws **24a-24d** may be loaded on the laminating stage **24** similarly to a drive source for driving and positioning the stage surface **24g** in the X-Y direction and the θ direction, but may be loaded outside the laminating stage **24** in order to reduce weight of the laminating stage **24**. In the case of loading the drive source of the claws **24a-24d** outside the laminating stage **24**, the external drive source and the claws **24a-24d** are connected by wires **24e** via a cam mechanism, drive force from the external drive source is transmitted by the wires **24e** and the claws **24a-24d** are swung and vertically driven. The claws **24a-24d** are pressurized in the vertically lower direction by a spring **24f**, and the 3-layer laminated body **30** and the 4-layer laminated body **40** on the stage surface **24g** are pressed and held by the pressurizing force.

(210) FIG. **37** illustrates a plane arrangement of the stage surface **24g** and the claws **24a-24d** of the laminating stage **24**. When the stage surface **24g** is in the rectangular shape, the claws **24a-24d** are arranged at the four corner parts of the rectangular shape. The claw **24a** and the claw **24c** are arranged on a diagonal line of the stage surface **24g**, and the claw **24b** and the claw **24d** are arranged on the other diagonal line of the stage surface **24g**. The claw **24a** and the claw **24c** are paired and are a first claw pair, and the claw **24b** and the claw **24d** are paired and are a second claw pair. The claws **24a-24d** are each freely swingable (freely turnable) clockwise and counterclockwise centering on the rotating shaft. When paying attention to the claw **24d**, it is arranged at an upper left corner of the stage surface **24g**, and the shaft is positioned within an area defined by extended lines of two opposite long sides of the stage surface **24g**. The claw **24d** is withdrawn to the outside (withdrawn position) of the existence area of the stage surface **24g** at the initial position, is partially positioned inside the existence area of the stage surface **24g** by swinging clockwise, and presses the laminated body (the 3-layer laminated body **30** and the 4-layer laminated body **40**) laminated on the stage surface **24g** at a pressing position. In addition, when paying attention to the claw **24a**, it is arranged at an upper right corner of the stage surface **24g**, and the shaft is positioned within the area defined by the extended lines of the two opposite long sides of the stage surface **24g**. The claw **24a** is withdrawn to the outside (withdrawn position) of the existence area of the stage surface **24g** at the initial position, is partially positioned inside the existence area of the stage surface **24g** by swinging counterclockwise, and presses and holds the laminated body laminated on the stage surface **24g**. The claw **24b** is operated the same as the claw **24d**, and the claw **24c** is operated the same as the claw **24a**.

(211) When pressing and holding the laminated body, the claw **24a** and the claw **24c**, and the claw **24b** and the claw **24d** positioned on the diagonal line of the stage surface **24g** are swung within the horizontal plane and also vertically moved respectively as sets. That is, the first claw pair of the claw **24a** and the claw **24c** is swung and presses and holds the laminated body, but the second claw pair of the claw **24b** and the claw **24d** does not hold the laminated body which is newly laminated since it is pressing and holding the laminated body which is already laminated at the time. Then, when the new laminated body is laminated next, the second claw pair of the claw **24b** and the claw **24d** is swung, withdrawn from the stage surface **24g**, moved in the upper direction and swung again, and presses and holds the laminated body which is newly laminated.

(212) FIG. **38** illustrates the operation of pressing and holding the laminated body by the claws **24a-24d**.

(213) FIG. **38(a)** illustrates the state of pressing and holding a 3-layer laminated body **30-1** laminated on the stage surface **24g**. When the first 3-layer laminated body **30-1** is laminated on the stage surface **24g**, the claw **24a** and the claw **24c** are swung counterclockwise and arranged on the 3-layer laminated body **30-1**, press the 3-layer laminated body **30-1** in the vertically lower direction from above by the pressurizing force of the spring **24f**, and hold the 3-layer laminated body **30-1**.

(214) FIG. **38(b)** illustrates the state where a new 4-layer laminated body **40-2** is laminated on the 3-layer laminated body **30-1** newly from the state of FIG. **38(a)**. Since the 4-layer laminated body **40-2** is laminated over the claws **24a** and **24c** pressing and holding the 3-layer laminated body **30-1**, the 4-layer laminated body **40-2** is laminated so as to cover a part of the claws **24a** and **24c**.

(215) FIG. **38(c)** illustrates the state of pressing and holding the 4-layer laminated body **40-2**. While the 3-layer laminated body **30-1** is kept pressed and held by the claws **24a** and **24c**, the claws **24b** and **24d** are moved in the vertically upper direction, swung clockwise further, and arranged on the 4-layer laminated body **40-2**. The claws **24b** and **24d** press the 4-layer laminated body **40-2** in the vertically lower direction from above by the pressurizing force of the spring **24f** and hold it.

(216) FIG. **38(d)** illustrates the state where the claws **24a** and **24c** are withdrawn from the state of FIG. **38(c)**. After the 4-layer laminated body **40-2** is pressed and held by the claws **24b** and **24d**, the claws **24a** and **24c** no longer need to press and hold the 3-layer laminated body **30-1**, and need to be withdrawn from the stage surface **24g** so as not to obstruct the laminated body to be newly laminated next. Then, the claws **24a** and **24c** are moved in the vertically upper direction, swung clockwise and withdrawn from the stage surface **24g**. At the time, since the 3-layer laminated body **30-1** and the 4-layer laminated body **40-2** are pressed and held by the claws **24b** and **24d**, the laminating position is not shifted by the swing of the claws **24a** and **24c**.

(217) FIG. **38(e)** illustrates the state where a next laminated body **40-3** is laminated on the 4-layer laminated body **40-2** newly from the state of FIG. **38(d)**. Since the 4-layer laminated body **40-3** is laminated over the claws **24b** and **24d** pressing and holding the 4-layer laminated body **40-2**, the 4-layer laminated body **40-3** is laminated so as to cover a part of the claws **24b** and **24d**.

(218) FIG. **38(f)** illustrates the state of pressing and holding the 4-layer laminated body **40-3**. While the 4-layer laminated body **40-2** is kept pressed and held by the claws **24b** and **24d**, the claws **24a** and **24c** are moved in the vertically upper direction, swung counterclockwise further, and arranged on the 4-layer laminated body **40-3**. The claws **24a** and **24c** press the 4-layer laminated body **40-3** in the vertically lower direction from above by the pressurizing force of the spring **24f** and hold it. By repeating the above-described operation, the 4-layer laminated bodies **40-2**, **40-3**, . . . are laminated successively on the 3-layer laminated body **30-1**.

(219) The claws **24a-24d** and the respective shafts in the present embodiment are arranged within the area defined by the extended lines of the two opposite long sides of the stage surface **24g** as illustrated in FIG. **37**, however, without being limited thereto, they may be arranged outside the area defined by the extended lines of the two opposite long sides of the stage surface **24g**. The inside of the area defined by the extended lines of the two opposite long sides of the stage surface

24g is the area including also the stage surface **24g**. The outside of the area defined by the extended lines of the two opposite long sides of the stage surface **24g** is the area not including the stage surface **24g**.

(220) FIG. **39** illustrates a plan view in the case where the respective shafts of the claws **24a-24d** are arranged outside the area defined by the extended lines of the two opposite long sides of the stage surface **24g**. When paying attention to the claw **24d**, it is arranged at the upper left corner of the stage surface **24g**, and the shaft is positioned outside the area defined by the extended lines of the two opposite long sides of the stage surface **24g**. The attention should be paid to a difference between the position of the claw **24d** in FIG. **37** and the position of the claw **24d** in FIG. **39**. It is similar for the other claws **24a-24c**. Note that, in FIG. **39**, the respective shafts of the claws **24a-24d** are positioned outside the area defined by the extended lines of two opposite short sides of the stage surface **24g**. The outside of the area defined by the extended lines of the two opposite short sides of the stage surface **24g** is the area not including the stage surface **24g**.

(221) When the respective shafts of the claws **24a-24d** are arranged outside the area defined by the extended lines of the two opposite long sides of the stage surface **24g**, in the case where the laminated body sucked and held on the outer peripheral surface of the laminating heads **22a-22m** of the laminating drum **22** is laminated onto the stage surface **24g**, the laminated body does not enter from right above the stage surface **24g**, and the laminated body enters from diagonally above the stage surface **24g** since the outer peripheral surface is moved along the circumference. When the radius of the laminating drum **22** is sufficiently large relatively compared to the size of the stage surface **24g**, it is almost the same as the entry of the laminated body from a side direction of the stage surface **24g**.

(222) FIG. **39** schematically illustrates the situation where the laminated body enters from the side direction of the stage surface **24g** in such a manner. The 4-layer laminated body **40-2** is laminated on the stage surface **24g**, and the 4-layer laminated body **40-2** is pressed and held by the claws **24a** and **24c**. In the case of newly laminating the next 4-layer laminated body **40-3**, the 4-layer laminated body **40-3** sucked and held on the outer peripheral surface of the laminating drum **22** enters from the side (from below in the figure) of the stage surface **24g**. The shafts of the claws **24b** and **24d** are arranged outside the area defined by the extended lines of the two opposite long sides of the stage surface **24g**. The claws **24b** and **24d** are moved in the vertically upper direction and swung, and do not collide with the next 4-layer laminated body **40-3** entering from the side even when they are positioned on the stage surface **24g**. The claws **24b** and **24d** are not obstacles to the 4-layer laminated body **40-3**. Accordingly, when laminating the 4-layer laminated body **40-3**, the claws **24b** and **24d** do not need to be withdrawn to the outside (withdrawn position) of the existence area of the stage surface **24g**. The claws **24b** and **24d** are positioned on the stage surface **24g** (standby position) and made to stand by, and the claws **24b** and **24d** can press and hold the 4-layer laminated body **40-3** by the pressurizing force of the spring **24f** promptly after the 4-layer laminated body **40-3** is laminated on the 4-layer laminated body **40-2**.

(223) FIG. **40-FIG. 44** illustrate the operation of the claws **24a-24d** of the laminating stage **24** linked with the movement of the laminating heads **22a-22m** in more detail. In the figures, the movement of the laminating head **22a** among the laminating heads **22a-22m** is illustrated, however, it is also similar for the other laminating heads **22b-22m**.

(224) FIG. **40** illustrates a side view of the case where the 3-layer laminated body **30-1** and the 4-layer laminated body **40-2** are already laminated on the stage surface **24g** of the laminating stage **24** and the 4-layer laminated body **40-3** is to be newly laminated next by the laminating head **22a**. While the figure illustrates the claws **24c** and **24d**, the claw **24a** is operated similarly to the claw **24c**, and the claw **24b** is operated similarly to the claw **24d**.

(225) The claw **24c** presses and holds the 3-layer laminated body **30-1** and the 4-layer laminated body **40-2** on the stage surface **24g** by the pressurizing force of the spring **24f**. On the other hand, the claw **24d** is moved in the vertically upper direction and swung around the shaft and is in a

standby state so as to be partially positioned above the stage surface **24g** (withdrawn position). The laminating head **22a** is rotated while sucking and holding the 4-layer laminated body **40-3** on the outer peripheral surface, is accelerated when the specific position is reached, and approaches the position close to the stage surface **24g**. When the laminating head **22a** approaches the position close to the stage surface **24g**, the holding portion **22a3** is swung to the arms **22a1** and **22a2** and the outer peripheral surface is made roughly parallel to the stage surface **24g** while avoiding a collision with the stage surface **24g**.

(226) FIG. **41** is the case where the laminating head **22a** reaches the position close to the stage surface **24g**.

(227) The arms **22a1** and **22a2** of the laminating head **22a** are moved in the radial direction of the laminating drum **22**, the holding portion **22a3** is also moved in the vertically lower direction, and the outer peripheral surface of the holding portion **22a3** further gets closer to the stage surface **24g**. At the time, the claw **24d** is in the standby state above the stage surface **24g** (withdrawn position), and the claw **24c** maintains the state of pressing and holding the 3-layer laminated body **30-1** and the 4-layer laminated body **40-2**. Then, from the state, as illustrated in FIG. **42**, the laminating head **22a** laminates the 4-layer laminated body **40-3** onto the 3-layer laminated body **30-1** and the 4-layer laminated body **40-2** on the stage surface **24g**, and turns off the suction force of the holding portion **22a3**. The movement in the vertically lower direction of the holding portion **22a3** may be temporarily stopped after the movement in the vertically lower direction of the holding portion **22a3** is started and before the 4-layer laminated body **40-3** is laminated on the 3-layer laminated body **30-1** and the 4-layer laminated body **40-2** on the stage surface **24g**. By the temporary stop, an impact when the 4-layer laminated body **40-3** is laminated on the 3-layer laminated body **30-1** and the 4-layer laminated body **40-2** on the stage surface **24g** can be mitigated.

(228) FIG. **43** illustrates the state where the 4-layer laminated body **40-3** is laminated on the 3-layer laminated body **30-1** and the 4-layer laminated body **40-2**. Since the claw **24c** (and the claw **24a**) presses and holds the 3-layer laminated body **30-1** and the 4-layer laminated body **40-2**, the 4-layer laminated body **40-3** is laminated on the claw **24c** (and the claw **24a**). In addition, the claw **24d** (and the claw **24b**) is in the standby state above the stage surface **24g**. When the 4-layer laminated body **40-3** is laminated, the claw **24d** (and the claw **24b**) is promptly moved in the vertically lower direction, and presses and holds the newly laminated 4-layer laminated body **40-3**. When the 4-layer laminated body **40-3** is pressed and held by the claw **24d** (and the claw **24b**), since there is no need to press and hold the 3-layer laminated body **30-1** and the 4-layer laminated body **40-2** anymore, the claw **24c** (and the claw **24a**) is moved in the vertically upper direction, then swung around the shaft and withdrawn from the stage surface **24g**, and stands by above the stage surface **24g** (withdrawn position).

(229) FIG. **44** illustrates the state where the lamination of the 4-layer laminated body **40-3** is ended. The 3-layer laminated body **30-1**, the 4-layer laminated body **40-2** and the 4-layer laminated body **40-3** are pressed and held by the claw **24d** (and the claw **24b**). In addition, the claw **24c** (and the claw **24a**) is positioned above the stage surface **24g** (withdrawn position) in the standby state, and prepares for the lamination of the 4-layer laminated body to be newly laminated next, that is the 4-layer laminated body sucked and held on the outer peripheral surface of the laminating head **22b**.

(230) In such a manner, according to the arrangement of the claws **24a-24d** illustrated in FIG. **39**, since the two claws other than the claws pressing and holding the laminated body can stand by above the stage surface **24g** (withdrawn position) without the need of being withdrawn so as not to be obstructive when the next laminated body enters, the laminated body is promptly pressed and held after it is laminated and the total lamination time can be shortened.

(231) Note that the outer peripheral surface (holding portion **22a3**) of the laminating head **22a** may be configured such that the cross section viewed from the circumferential direction of the laminating drum **22** is not flat and is curved in a concave shape or a convex shape and the 3-layer laminated body or the 4-layer laminated body is sucked and held on the concave or the convex. It is

to suppress fluttering of the part of the 3-layer laminated body or the 4-layer laminated body protruding from the end of the laminating head **22a** even in the case where the size of the outer peripheral surface of the laminating head **22a** is made one size smaller than the size of the 3-layer laminated body or the 4-layer laminated body. In the case where the holding portion **22a3** of the laminating head **22a** is curved in the convex shape, the close position of the laminating head **22a** and the stage surface **24g** means the position at which the distance between the most convex part of the holding portion **22a3** and the stage surface **24g** is the shortest. In the case where the outer peripheral surface of the laminating head **22a** is curved in the concave shape, the close position of the laminating head **22a** and the stage surface **24g** means the position at which the distance between the part positioned at the outermost periphery of the laminating head **22a** and the stage surface **24g** is the shortest.

(232) The embodiment of the present disclosure has been described above, and the present disclosure is not limited to the embodiment and can be various modified.

(233) Hereinafter, modifications will be described.

(234) <Modification 1>

(235) While only one set of the laminating drum **22** and the laminating stage **24** is arranged in the present embodiment, the plurality of sets of the laminating drum **22** and the laminating stage **24** may be arranged.

(236) FIG. **45** illustrates a configuration diagram in the case of arranging a laminating drum **23** and a laminating stage **25** in addition to the laminating drum **22** and the laminating stage **24**.

(237) The 3-layer laminated body **30** and the 4-layer laminated body **40** generated by being cut at the separator cutting drum **20** are supplied to the laminating drum **22** or the laminating drum **23**. The 3-layer laminated body **30** and the 4-layer laminated body **40** supplied to the laminating drum **22** are successively laminated on the laminating stage **24** arranged adjacently to the laminating drum **22**. In addition, the 3-layer laminated body **30** and the 4-layer laminated body **40** supplied to the laminating drum **23** are successively laminated on the laminating stage **25** arranged adjacently to the laminating drum **23**.

(238) The laminating drum **22** and the laminating drum **23** are each in contact with the separator cutting drum **20**, and are mutually rotated in the same direction as the rotation direction of the separator cutting drum **20**. The arrangement that the plurality of laminating drums **22** and **23** are rotated in the same direction as a conveyance direction of the 3-layer laminated body **30** and the 4-layer laminated body **40** conveyed from the separator cutting drum **20** in such a manner is referred to as a parallel arrangement. In the parallel arrangement, structures of the laminated bodies laminated on the laminating stages **24** and **25** are identical.

(239) FIG. **46** illustrates the situation of the lamination using the set of the laminating drum **22** and the laminating stage **24** and the set of the laminating drum **23** and the laminating stage **25**.

(240) The controller which controls the entire manufacturing device selectively supplies the 3-layer laminated body **30** and the 4-layer laminated body **40** generated by being cut at the separator cutting drum **20** to one of the laminating drum **22** and the laminating drum **23**. For example, in the case of first manufacturing the laminated electrode assembly by performing the lamination on the laminating stage **24** and then manufacturing the laminated electrode assembly by performing the lamination on the laminating stage **25**, the controller supplies the 3-layer laminated body **30** and the 4-layer laminated body **40** from the separator cutting drum **20** to the laminating drum **22** first.

(241) The laminating drum **22** receives the 3-layer laminated body **30** supplied from the separator cutting drum **20** and laminates it on the laminating stage **24**. Then, on the 3-layer laminated body **30**, the total of 37 pieces of the 4-layer laminated bodies **40** for example are laminated and one laminated electrode assembly is manufactured. When the last 37th 4-layer laminated body **40** is supplied to the laminating drum **22**, the controller switches an output destination of the separator cutting drum **20** from the laminating drum **22** to the laminating drum **23**.

(242) The laminating drum **23** receives the 3-layer laminated body **30** supplied from the separator

cutting drum **20** and laminates it on the laminating stage **25**. Then, on the 3-layer laminated body **30**, the total of 37 pieces of the 4-layer laminated bodies **40** for example are laminated and another laminated electrode assembly is manufactured. When the last 37th 4-layer laminated body **40** is supplied to the laminating drum **23**, the controller switches the output destination of the separator cutting drum **20** from the laminating drum **23** to the laminating drum **22** again.

(243) There is no problem when the 3-layer laminated body **30**, the 4-layer laminated body **40**, the 4-layer laminated body **40**, . . . , the 3-layer laminated body **30**, the 4-layer laminated body **40**, . . . are normally supplied from the separator cutting drum **20**, but there may be a case where abnormality occurs in at least one of the negative electrode cutting drum **10**, the negative electrode heating drum **12**, the positive electrode cutting drum **14**, the positive electrode heating drum **16**, the bonding drum **18** and the separator cutting drum **20** and normal 3-layer laminated body **30** and 4-layer laminated body **40** are not generated. In such a case, when the abnormality is detected by a detection sensor provided in each drum, the 3-layer laminated body **30** or the 4-layer laminated body **40** is eliminated as a defect before being supplied to the laminating drums **22** and **23**, and a normal flow of the laminated bodies that 37 pieces of the 4-layer laminated bodies **40** follow after one 3-layer laminated body **30** is not maintained. For example, at the time of generating the 37th 4-layer laminated body **40**, when a bonding defect of the positive electrode plate PP is generated at the bonding drum **18** and thus the 37th 4-layer laminated body **40** is determined as a defect and eliminated in an inspection process of the bonding drum **18**, not the 4-layer laminated body **40** but the next 3-layer laminated body **30** is supplied for the 37th piece from the separator cutting drum **20**. When the 3-layer laminated body **30** is laminated on the laminating stage **24** from the laminating drum **22**, the normal laminated electrode assembly is not manufactured.

(244) Then, in such a case, it is also possible to supply the 3-layer laminated body **30** supplied instead of the 37th 4-layer laminated body **40** not to the laminating drum **22** but to the laminating drum **23** and make the laminating drum **23** laminate the 3-layer laminated body **30** on the laminating stage **25**. By supplying the 4-layer laminated body **40** supplied from the separator cutting drum **20** after the 3-layer laminated body to the laminating drum **22** and laminating it on the laminating stage **24**, the laminated body for which the 37 pieces of the 4-layer laminated bodies **40** are laminated on the 3-layer laminated body **30** is formed on the laminating stage **24**.

(245) Note that the laminated body for which the predetermined number of the 4-layer laminated bodies **40** are laminated on the 3-layer laminated body **30** is pressurized and/or heated, bonded to each other and turned to the laminated electrode assembly.

(246) FIG. **47** schematically illustrates the lamination method in this case.

(247) The controller monitors the laminated body generated by being cut at the separator cutting drum **20**, identifies whether it is the 3-layer laminated body **30** or the 4-layer laminated body **40** and also successively counts the number of pieces. When the abnormality occurs in the 37th 4-layer laminated body **40** and it is eliminated as a defect, the 37th piece becomes not the 4-layer laminated body **40** but the 3-layer laminated body **30**.

(248) When the 37th laminated body is the 4-layer laminated body **40**, the controller supplies it to the laminating drum **22**, however, when it is detected that the 37th laminated body is not the 4-layer laminated body **40** but the 3-layer laminated body **30**, changes the output destination of the separator cutting drum **20** from the laminating drum **22** to the laminating drum **23** and laminates the 3-layer laminated body **30** not on the laminating stage **24** but on the laminating stage **25**. In the figure, an X mark attached to the 3-layer laminated body **30** indicates that it is not to be laminated on the laminating stage **24** and is to be laminated on the laminating stage **25**. Thereafter, since the 4-layer laminated bodies **40** are successively supplied, the controller maintains the output destination of the separator cutting drum **20** as the laminating drum **23**, and successively laminates the 37 pieces of the 4-layer laminated bodies **40** on the laminating stage **25**. For the laminating stage **24**, by laminating one 4-layer laminated body **40** in need at an appropriate timing, the laminated electrode assembly is completed.

(249) By arranging the plurality of sets of the laminating drum and the laminating stage and appropriately distributing the laminated bodies, even when a defect occurs in the laminated body, the laminated electrode assembly can be efficiently manufactured without stopping the manufacturing process.

(250) While the defect of the laminated body can be detected at any drum, in order to detect the defect of the negative electrode plate NP and the defect of the positive electrode plate PP in particular, the detection sensor such as a camera is arranged at the negative electrode heating drum **12** and the positive electrode heating drum **16** and the defect detected at the drums can be eliminated. In this case, the negative electrode heating drum **12** comprises a normal/defect determination function and a defect elimination function together with a function of heating the negative electrode plate NP. Similarly, the positive electrode heating drum **16** comprises the normal/defect determination function and the defect elimination function together with a function of heating the positive electrode plate PP.

(251) In the case of detecting the defect on the negative electrode cutting drum **10** and the positive electrode cutting drum **14**, after the negative electrode single plate or the positive electrode single plate is cut, the normal/defect determination function and the defect elimination function are provided. The negative electrode cutting drum **10** and the positive electrode cutting drum **14** may comprise the normal/defect determination function and the negative electrode heating drum **12** and the positive electrode heating drum **16** may comprise the defect elimination function.

(252) In the case of detecting the defect on the separator cutting drum **20**, after the separator is cut and before the cut laminated body is supplied to the laminating drum, the normal/defect determination function and the defect elimination function are provided. That is, the separator cutting drum **20** comprises the normal/defect determination function and the defect elimination function together with a function of cutting the separator.

(253) <Modification 2>

(254) While the laminating drums **22** and **23** are arranged in parallel in FIG. **45**, the laminating drums **22** and **23** may be arranged in series to the separator cutting drum **20**.

(255) FIG. **48** illustrates the configuration of this case. The 3-layer laminated body **30** and the 4-layer laminated body **40** generated by being cut at the separator cutting drum **20** are supplied to the laminating drum **22**. The laminating drum **23** is arranged in contact with not the separator cutting drum **20** but the laminating drum **22**. To the laminating drum **23**, the 3-layer laminated body **30** and the 4-layer laminated body **40** are supplied via the laminating drum **22**. The 3-layer laminated body **30** and the 4-layer laminated body **40** supplied to the laminating drum **22** are successively laminated on the laminating stage **24** arranged adjacently to the laminating drum **22**. In addition, the 3-layer laminated body **30** and the 4-layer laminated body **40** supplied to the laminating drum **23** are successively laminated on the laminating stage **25** arranged adjacently to the laminating drum **23**.

(256) The laminating drum **22** and the laminating drum **23** are rotated in the mutually different directions. The arrangement that the laminating drums **22** and **23** in contact are rotated in the mutually different directions in such a manner is referred to as a series arrangement. The rotation in the mutually different directions means the case where one drum is rotated clockwise and the other drum is rotated counterclockwise. In the series arrangement, the structures of the laminated bodies laminated on the laminating stages **24** and **25** may be different.

(257) That is, the laminating drum **23** is supplied with the 3-layer laminated body **30** and the 4-layer laminated body **40** via the laminating drum **22** in contact, but the rotation direction of the laminating drum **23** is opposite to the rotation direction of the laminating drum **22**. The 3-layer laminated body **30** and the 4-layer laminated body **40** supplied from the laminating drum **23** to the laminating stage **25** are inverted in regard to the arrangement of the 3-layer laminated body **30** and the 4-layer laminated body **40** supplied from the laminating drum **22** to the laminating stage **24**. Then, for example, while the 3-layer laminated body is laminated first and the 4-layer laminated

bodies are successively laminated on the 3-layer laminated body on the laminating stage **24**, the 4-layer laminated bodies are successively laminated and the 3-layer laminated body is laminated on the 4-layer laminated body on the laminating stage **25**. Accordingly, in the series arrangement, the structures of the laminated bodies laminated on the laminating stages **24** and **25** may be different. (258) Note that, when the configuration is such that the laminating drum **23** is not brought into contact with the laminating drum **22** but an intermediate drum **27** is interposed to the laminating drum **22** and the laminating drum **23** is brought into contact with the intermediate drum **27** as illustrated in FIG. **49**, the laminating drum **22** and the laminating drum **23** are rotated in the same direction to the separator cutting drum **20** so that it is the parallel arrangement.

(259) <Modification 3>

(260) While the present embodiment is the configuration of the negative electrode cutting drum **10**, the negative electrode heating drum **12**, the positive electrode cutting drum **14**, the positive electrode heating drum **16**, the bonding drum **18**, the separator cutting drum **20** and the laminating drum **22**, the negative electrode heating drum **12** may be omitted to adjacently arrange the negative electrode cutting drum **10** and the bonding drum **18** and the positive electrode heating drum **16** may be omitted to adjacently arrange the positive electrode cutting drum **14** and the bonding drum **18**.

(261) FIG. **50** illustrates the configuration of this case. Differently from FIG. **1**, the negative electrode heating drum **12** and the positive electrode heating drum **16** are not provided, the negative electrode cutting drum **10** and the bonding drum **18** are adjacent and the positive electrode cutting drum **14** and the bonding drum **18** are adjacent.

(262) The negative electrode cutting drum **10** generates the negative electrode plate NP by cutting the belt-like negative electrode single plate N, heats the negative electrode plate NP and then supplies it to the bonding drum **18**. It can be said that the negative electrode cutting drum **10** also has the function of the negative electrode heating drum **12** in FIG. **1**.

(263) In addition, the positive electrode cutting drum **14** generates the positive electrode plate PP by cutting the belt-like positive electrode single plate P, heats the positive electrode plate PP and then supplies it to the bonding drum **18**. It can be said that the positive electrode cutting drum **14** also has the function of the positive electrode heating drum **16** in FIG. **1**.

(264) The negative electrode cutting drum **10** and the bonding drum **18** are adjacent in FIG. **50**, however, the negative electrode cutting drum **10** and the bonding drum **18** may be separated and the heated negative electrode plate NP from the negative electrode cutting drum **10** may be conveyed to the bonding drum **18** by a conveyance mechanism such as a belt conveyor. Similarly, the positive electrode cutting drum **14** and the bonding drum **18** may be separated and the heated positive electrode plate PP from the positive electrode cutting drum **14** may be conveyed to the bonding drum **18** by a conveyance mechanism such as a belt conveyor.

(265) <Modification 4>

(266) While the belt-like separators S1 and S2 are cut at the separator cutting drum **20** in the present embodiment, the separator cutting drum **20** may be omitted and the belt-like separators S1 and S2 may be cut at the bonding drum **18**.

(267) FIG. **51** illustrates the configuration of this case. Differently from FIG. **1**, the separator cutting drum **20** is not provided and the 3-layer laminated body **30** and the 4-layer laminated body **40** generated by being cut at the bonding drum **18** are supplied to the laminating drum **22** via the conveyance mechanism.

(268) The bonding drum **18** comprises a bonding head comprising a round blade, and cuts the belt-like separators S1 and S2 at a predetermined cutting position **28**. The bonding head may comprise the round blade and a cutting function for example.

(269) <Modification 5>

(270) The 3-layer laminated body and the 4-layer laminated body are created in the present embodiment, however, without being limited thereto, a 2-layer laminated body or the laminated body with the number of laminations equal to or larger than four can be also created. For example,

the positive electrode cutting drum **14** and the positive electrode heating drum **16** may be eliminated or stopped in FIG. **1** and the negative electrode plate NP may be supplied onto the separator **S1**, bonded at the bonding drum **18** and supplied to the separator cutting drum **20** as the 2-layer laminated body. The separator **S2** may be supplied onto the negative electrode plate NP, bonded at the bonding drum **18** and supplied to the separator cutting drum **20** as the 2-layer laminated body. The separator **S2** may be supplied onto the negative electrode plate NP and bonded at the bonding drum **18** and the positive electrode plate PP may be supplied further onto the separator **S2**, bonded at the bonding drum **18** and supplied to the separator cutting drum **20** as the 3-layer laminated body. Or, the laminated body with the number of laminations equal to or larger than four may be created and supplied to the separator cutting drum **20** by repeatedly executing the bonding at the bonding drum **18**.

(271) <Modification 6>

(272) The separator cutting drum **20** and the laminating drum **22** do not need to be arranged closely to the bonding drum **18**. The laminated body may be created by laminating the belt-like separator, the positive electrode and the negative electrode by the method and the device different from the bonding drum **18**, the belt-like separator may be cut and it may be laminated at the laminating drum **22**. On the other hand, the laminated body for which the belt-like separator, the positive electrode and the negative electrode are laminated may be created at the bonding drum **18** and the laminated electrode assembly may be created by the method and the device different from the separator cutting drum **20** and the laminating drum **22**.

REFERENCE SIGNS LIST

(273) **10** negative electrode cutting drum **12** negative electrode heating drum **14** positive electrode cutting drum **16** positive electrode heating drum **18** bonding drum **20** separator cutting drum **22** laminating drum **24** laminating stage N negative electrode single plate P positive electrode single plate NP negative electrode plate PP positive electrode plate

Claims

1. An electrode assembly cutting device, comprising: a plurality of holding heads arranged in a circumferential direction of a drum and rotated around a drum center axis, the plurality of holding heads configured to hold a rectangular electrode assembly; and a plurality of cutting means, each of the plurality of the cutting means provided in a corresponding one of the plurality of holding heads and configured to cut the rectangular electrode assembly in a predetermined width, wherein said each of the plurality of cutting means comprises a pair of blades arranged in a respective radial direction of the drum so as to hold the rectangular electrode assembly between each of the pairs of blades.
2. The cutting device according to claim 1, wherein each of the pairs of blades is configured to cut the rectangular electrode assembly while maintaining a distance so as not to overlap with each other in the radial direction of the drum.
3. The electrode assembly cutting device according to claim 2, wherein a distance L of each of the pairs of blades in the respective radial direction of the drum greater than 0 μm and less than 250 μm .
4. The electrode assembly cutting device according to claim 3, wherein the distance L of each of the pairs of blades in the respective radial direction of the drum greater than 0 μm and less than 50 μm .
5. The electrode assembly cutting device according to claim 4, wherein the electrode assembly cutting device is configured to cut the rectangular electrode assembly by tensile stress by rotation of the holding heads.
6. The electrode assembly cutting device according to claim 2, wherein the plurality of holding heads are independently movable in the circumferential direction, so as to exert a mutually reverse

tensile force on the rectangular electrode assembly.

7. The electrode assembly cutting device according to claim 1, wherein said each of the plurality of cutting means is configured to cut the rectangular electrode assembly while making a forward movement and a return movement sequentially in a direction orthogonal to the circumferential direction of the drum.
8. The electrode assembly cutting device according to claim 7, wherein a respective forward movement path along which said each of the plurality of cutting means travels during the forward movement and a respective return movement path along which said each of the plurality of cutting means travels during the return movement are different from each other.
9. The electrode assembly cutting device according to claim 8, wherein said each of the plurality of cutting means is configured to form a cut between a first part and a second part of the rectangular electrode assembly during the forward movement, the first part and the second part of the rectangular electrode assembly being located on a first holding head and a second holding head, respectively, of the plurality of holding heads adjacent to each other, the first holding head is configured to hold the first part of the rectangular electrode assembly while moving away from the second holding head in the circumferential direction of the drum, and after forming the cut, said each of the plurality of cutting means is configured to move away from the rectangular electrode assembly, and thereafter make the return movement.
10. The electrode assembly cutting device according to claim 7, wherein each blade of each of the pairs of blades is a rotary round blade.
11. The electrode assembly cutting device according to claim 7, wherein each blade of each of the pairs of blades is a rotary round blade with a flat portion formed at one part.
12. The electrode assembly cutting device according to claim 11, wherein each of the pairs of blades is configured to cut the rectangular electrode assembly by round blade portions facing each other during the forward movement, and each of the pairs of blades is configured to remain apart from the rectangular electrode assembly while the flat portions face each other during the return movement.
13. A separator cutting device, comprising: a plurality of holding heads arranged in a circumferential direction of a drum and rotated around a drum center axis, the plurality of holding heads configured to hold a separator; and a plurality of cutting means, each of the plurality of the cutting means provided in a corresponding one of the plurality of holding heads and configured to cut the separator in a predetermined width, wherein said each of the plurality of cutting means comprises a pair of blades arranged in a respective radial direction of the drum so as to hold the separator between each of the pairs of blades, and said each of the plurality of cutting means is configured to cut the separator while making a forward movement and a return movement sequentially in a direction orthogonal to the circumferential direction of the drum.
14. The separator cutting device according to claim 13, wherein a respective forward movement path along which said each of the plurality of cutting means travels during the forward movement and a respective return movement path along which said each of the plurality of cutting means travels during the return movement are different from each other.
15. The separator cutting device according to claim 14, wherein said each of the plurality of cutting means is configured to form a cut between a first part and a second part of the separator during the forward movement, the first part and the second part of the separator being located on a first holding head and a second holding head, respectively, of the plurality of holding heads adjacent to each other, the first holding head is configured to hold the first part of the separator while moving away from the second holding head in the circumferential direction of the drum, and after forming the cut, said each of the plurality of cutting means is configured to move away from the separator, and thereafter make the return movement.
16. The separator cutting device according to claim 13, wherein each blade of each of the pairs of blades is a rotary round blade.

17. The separator cutting device according to claim 16, wherein each blade of each of the pairs of blades is a rotary round blade with a flat portion formed at one part.

18. The separator cutting device according to claim 17, wherein each of the pairs of blades is configured to cut the separator by round blade portions facing each other during the forward movement, and each of the pairs of blades is configured to remain apart from the separator while the flat portions face each other during the return movement.
